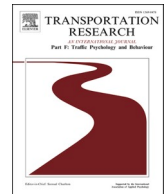




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Cognitive load and task switching in drivers: Implications for road safety in semi-autonomous vehicles

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ABSTRACT

As the semi-automation of motor vehicles advances, the prevalence of multitasking and task switching while driving has increased. In the next phase, known as conditional automated driving (level 3 automation), drivers will be able to fully engage in distracting tasks, yet they must be prepared to promptly resume control of the vehicle and maintain safe driving if requested to by the vehicle. In such situations, the driver's ability to flawlessly switch between the distracting task and the driving task becomes vitally important. This narrative review discusses conditional automated driving within the framework of cognitive psychology concepts of attention and task switching. Delayed reaction time and deteriorated driving performance are attributed to cognitive overload and switch cost. Factors that contribute to driving switch cost are identified and categorized, and several road safety concerns are raised, including: i) switch cost may last for between 20 s to 5 min; ii) inexperienced drivers may be unable to adequately control the vehicle after resuming manual control; and iii) low- and high-intensity non-driving tasks have a greater impact on takeover performance. To minimise the risk to road safety, suggestions have been provided to vehicle manufacturers, road users and regulatory authorities.

1. Introduction

Over the past few decades, motor vehicles have become increasingly sophisticated, largely with the aim of making the driving experience more enjoyable and safer. Technical advancements include split-zone air conditioning, cruise control, satellite navigation, mobile phone pairing, lane control monitoring, vehicle proximity monitoring, vehicle status monitoring and heads-up displays. These advancements increase the level of driver interaction with their vehicle, resulting in increased active involvement in operating or controlling the vehicle's functions, as well as more engagement in non-driving activities. Researchers have examined the impact of

Abbreviations: ADS, Automated Driving Systems; DDT, dynamic driving task; DMS, Driver monitor system; DRT, Detection-response Task; ISO, The International Organization for Standardization; LCT, Lane Change Task; LTM, long term memory; MRT, Multiple Resource Theory; NDRT, non-driving related task; RT, Reaction Time; SAE, Society of Automotive Engineers; SDLP, standard deviation of lateral position; SuRT, Surrogate Reference task; WM, Working memory; UNECE, The United Nations Economic Commission for Europe.

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some of these activities on driving performance. A review of various forms of communication during driving, including talking on a mobile phone, conversing with a passenger, and dialling while driving, concluded that all these forms of communication led to a decline in driving performance, with dialling having the most significant impact in terms of performance cost (Caird et al., 2018). Other studies have reported that texting while driving has negative effects on driving performance (Voinea et al., 2023). Radio tuning while driving increases the crash risk (Lee et al., 2018), and interactions with the central displays of the human–machine interface system in vehicles can also affect drivers' attention (Ma et al., 2020). In the United States in 2019, more than 3,100 deaths and 424,000 injuries involved distracted driving (CDC, 2022). In many countries, it is an offence for drivers to switch their attention to other activities, such as mobile phone use, while driving. Nonetheless, the increasingly complex nature of motor vehicles requires drivers to periodically switch their attention to the vehicle's controls.

Since the Society of Automotive Engineers (SAE) launched the standard for levels of driving automation (SAE, 2021) in 2014, vehicle automation has progressed dramatically. Mercedes-Benz was one of the first automotive manufacturers to release a level 3 automated vehicle onto the market in Germany in 2022 and in the US (Mercedes-Benz, 2023), with other manufacturers expected to follow. Level 3 automation introduces a new set of concerns regarding driver distraction and the ability to successfully take over the vehicle and re-establish safe driving. To anticipate potential safety issues arising from level 3 automation, researchers have investigated scenarios for this level of automation and evaluated driver performance (Eriksson & Stanton, 2017; Gong, Guo, Qi, Liang, & Wang, 2023; Wu, Wu, Lyu, & Zheng, 2019; Zeeb, Buchner, & Schrauf, 2015, 2016). At the current levels of automated driving from level 0 to level 2, drivers are required to drive with their full attention. However, in level 3 conditional automated driving, drivers are permitted to fully engage in non-driving related task (NDRT) unless they are prompted to take control of the vehicle. Studies on NDRT have been extensively conducted (Hungund & Kumar Pradhan, 2023; Jaussein et al., 2021). NDRTs used in conditional automated driving research can be categorised into standardised tasks and naturalistic tasks. While naturalistic tasks resemble real driving scenarios, standardised tasks are cognitive tasks that may be adapted from psychological tests that have been widely used, such as the n-back task (Kane et al., 2007; Melnicuk et al., 2021). For standardised cognitive tasks, task and driver engagement levels can be aligned to a range by comparing effects of different cognitive tasks on driver performance. For instance, researchers may increase the difficulty of a cognitive task to enhance driver engagement, thereby facilitating the investigation of corresponding changes in driver performance (Choi et al., 2020; Lu et al., 2021).

Technological advances in motor vehicles are requiring a greater level of interaction with the vehicle's controls and displays, which frequently necessitates the driver to temporarily direct their attention away from the road. Mobile communication systems in vehicles can also distract the driver, and the advent of level 3 automated driving introduces an entirely new set of activities to distract the driver from attending to the road and traffic conditions. The interrelated concepts of driver attention, switching of attention and cognitive load can adversely affect driving performance and road safety, and technological progress is steadily increasing their impact. The purpose of the present review is to provide automotive engineers and designers with a resource with which to understand the cognitive factors that affect driver attention, as well as a review of the related literature for researchers who are interested in investigating distracted driving. It is hoped that this knowledge will lead to the development of safer and more user-friendly driving systems.

This review will address the following topics:

- Cognitive psychology concepts relevant to driving task switching: An overview of psychological frameworks related to attention and task switching.
- Review method: A review of recent publications in the psychological, ergonomics, and engineering fields concerning task switching and driver distraction.
- Cognitive load in driving: A discussion of methods for assessing cognitive load in both manual and conditional automated driving scenarios and the effect of cognitive load on manual and conditional automated driving.
- Task switching in driving: The application of task switching theory in driving and the main factors affecting the effectiveness of task transition.
- Discussion: We raise road safety concerns that are associated with driver task switching and offer suggestions for mitigating risks.

2. Cognitive psychology concepts relevant to driving task switching

This section will show how concepts developed by psychologists can provide an understanding of the cognitive processes that are brought into play when a driver switches their attention from one task to another, and the implications that these processes have for road safety, particularly in the context of semi-autonomous vehicles. Cognitive psychology encompasses a massive literature, characterised by competing concepts and terminologies, and sometimes conflicting findings. Since the current review is intended for an audience of non-psychologists, psychological concepts and terminologies are presented here in their simplest forms without a discussion of the nuances.

2.1. Cognitive process during driving

The psychological basis of cognitive changes while driving has its basis in the concept of cognitive load. This concept assumes that cognitive architecture includes *long-term memory* (LTM) and *working memory* (WM) (Sweller, 2011; Sweller & Sweller, 2006). LTM involves the retrieval into consciousness of previous experiences and can be strengthened through learning, practice, and repetition, often on a subconscious level (Skaugset et al., 2016). For the purpose of this review, WM can be considered to be synonymous with short-term memory and it can be conceptualized as a kind of workstation with limited capacity in which new and old information is

constantly being transformed, combined, and updated (Baddeley, 1992; Chase & Ericsson, 1982; Solso et al., 2005). Most people can retain up to 5–7 separate bits (or ‘chunks’) of information in WM, and if the requirement to store new information surpasses this capacity, some bits of information are erased, in a situation known as information overload. Cognitive load refers to the amount of information that WM is required to store and operate on at a given time. As the number of bits of information increases, the effort required to retain these bits in WM becomes greater (Atkinson & Shiffrin, 1968). Some researchers use the term “cognitive workload” rather than “cognitive load” (Cooper et al., 2013; Patten et al., 2006; von Janczewski et al., 2021). Cognitive workload is a similar term with a subtle difference: cognitive load is correlated with task difficulty whereas cognitive workload takes into account individual differences in ability (McKendrick & Harwood, 2019); the same cognitive load can result in different workload levels, as individuals have different maximum WM capacities. The present review refers only to cognitive load, as individual differences are not a focus of discussion. Driver behaviour studies might use the term cognitive load to refer to the mental demand imposed on the driver by a non-driving task, especially non-visual tasks (Goodridge et al., 2023). In the present paper, cognitive load includes both the driving task and the distracting task.

Undertaking a cognitive task requires *attention*. In cognitive psychology, attention is defined as “the concentration of mental effort on specific sensory or mental events” (Schneider & Shiffrin, 1977; Solso et al., 2005). In attention and cognitive control, human performance is hypothesised to be driven by two modes, automatic and controlled (Servant et al., 2018). *Automaticity* allows people to perform tasks fast, efficiently, unconsciously, and goal-unrelatedly (Servant et al., 2018) and involves little cognitive load. Automaticity theories posit that through repeated practice, the control of cognitive processes shifts from a reliance on WM to becoming automatic and guided by the retrieval processes of LTM. The main limitation of the automatic/involuntary process is inflexibility, which means the response can be inappropriate when conditions change (Eysenck & Keane, 2020). In contrast to automatic processes, non-automatic/voluntary processes require attention and engagement of WM (Shiffrin & Schneider, 1977), which require a cognitive load, though they can be used flexibly in changing circumstances (Eysenck & Keane, 2020).

Classically, the control of attention has been categorised into two paths: *top-down attention* (goal-directed) and *bottom-up* (stimulus-driven) (Anderson et al., 2021). When people are undertaking a task, such as driving, they must be able to maintain concentration on the task and resist interference from irrelevant information. ‘Top-down’ attention refers to search strategies that are consciously implemented by the individual (e.g. looking out for stop signs), while ‘bottom-up’ attention involves the perception of salient stimuli that involuntarily capture our attention (e.g. noticing flashing lights on a police car). While both forms of attention are important for driving, many bottom-up stimuli are irrelevant and can distract the driver. Attention is considered to be related to WM capacity (Kane & Engle, 2002; Oberauer, 2019). It has been proposed that people with a high WM capacity are generally better at maintaining top-down attentional control and inhibiting responses to bottom-up stimuli (Furley & Memmert, 2012; Hiebel & Zimmer, 2015). Recently a third path was recognised. Selection history, which involves learning and practice and is stored in LTM, has been shown to influence attention and responses to familiar situations (Anderson et al., 2021).

Fig. 1 summarises the involvement of attention and cognitive processing during driving. Cognitive processes and attention are important for safe driving: the driver must keep the car within the lane while constantly assessing new sensory information and taking

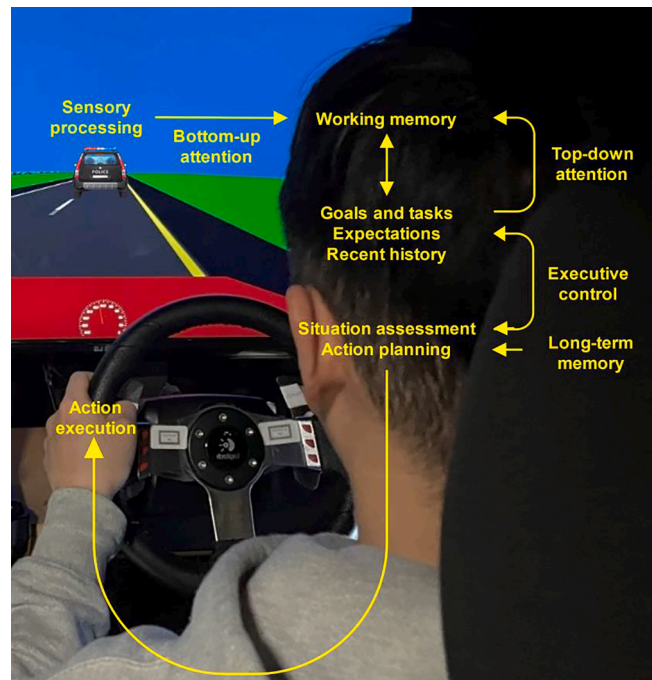


Fig. 1. Schematic overview of the roles of attention and cognitive processing in driving, modified from (Remington & Loft, 2015).

appropriate action as necessary until the destination is reached. The retention of sensory input (e.g. a police car ahead) in working memory is influenced by both bottom-up (e.g. noticing flashing lights) and top-down (e.g. maintaining the car within the lane) attentional processes and is modified by goals, tasks, expectations, and recent history. Information accumulates on more active goals until the driver has assessed the situation and selected a response. Executive control coordinates tasks and goals, including suspending and switching from one task to another. Situation assessment and action planning are also under executive control. Automatic responses stored in LTM assist with situation assessment and action planning that guide the execution of actions.

An important aspect of top-down attention is the extent to which available cognitive processes or energy are allocated to the task. As a person turns their attention to a task, the amount of attention given to that task can vary from a cursory involvement to an exclusive involvement. Someone who is fully focussed on a task does so to the exclusion of all other tasks and inhibits or blocks incoming stimuli that would normally capture their attention. Under this condition, the phenomenon of people failing to notice a salient and fully visible, but unexpected object or event is known as *inattention blindness*. (Kreitz, Furley, Memmert, & Simons, 2015). For example, when a novice driver fully engrossed in driving on a road with complex road conditions is asked a question, they might not respond because they were not consciously aware that a question had been asked. This capacity to concentrate deeply on a task, to the exclusion of other tasks and stimuli, has been variously termed *focussed attention*, sustained attention, attentional stability, and task focus. Focussed attention is influenced by factors such as motivation, drive, and arousal; it can be decreased by the passage of time and by external distractors, furthermore, individuals differ in their capacity for focused attention (Cohen, 2014). Focussed attention is counterbalanced by cognitive flexibility, which refers to the capacity to readily switch attention from one task to another in response to internal or external cues. If focused attention is not allocated, it may cause *change blindness*, a phenomenon where people fail to observe visual changes that are easily seen once noticed (Rensink, 2009). For example, drivers may fail to notice a road sign or a pedestrian, which happens more in urban scenes than in rural scenes (Beanland et al., 2017).

2.2. Task switching

The previous section introduced basic psychological concepts and terminology involved in driving, especially cognitive load and attention. Building on this, driving with a distracting task brings the challenge of multitasking. Multitasking, particularly in the context of driving, presents a significant challenge to performance and safety. Drivers who engage in secondary tasks while driving are effectively multitasking, which involves either performing tasks concurrently or alternating between them. Multitasking can be broadly categorized into two forms: concurrent multitasking, where tasks are performed simultaneously, and serial multitasking, which involves task-switching (MacPherson, 2018). The impact of multitasking on task performance has been extensively studied, particularly with regard to the allocation of cognitive resources. To elucidate the mechanisms underlying the observed performance decrements during multitasking, the Multiple Resource Theory (MRT) offers a framework for understanding how different tasks compete for limited cognitive resources. Wickens proposed a four-dimensional computational model that categorizes mental resources along the following dimensions: processing stages (cognitive vs. motor tasks), processing codes (spatial vs. verbal tasks), modalities (auditory vs. visual tasks), and visual channels (focal vs. ambient vision) (Wickens, 2008; Wickens, 2014). This multidimensional model clarifies why tasks that draw on the same mental resources are more likely to interfere with one another, thereby reducing overall performance. In the context of driving, the MRT model is particularly useful for understanding how the cognitive demands of operating a vehicle interact with the demands of engaging in secondary tasks, potentially compromising driver performance and safety.

As automation levels increase, drivers' multitasking is more likely to involve task switching, as they have greater opportunities to engage in non-driving tasks compared to lower levels of automation (e.g., Level 0 or Level 1). When drivers shift their attention between driving and these secondary tasks, this task switching can impact their attention and performance. Task switching frequently occurs in everyday life and is considered to be at the nexus of executive control in the human cognitive system (Schneider & Logan, 2009). Task switching prolongs reaction time (RT) compared to non-switching trials (Monsell, 2003). This is because when people switch attention between tasks, each cognitive task requires an appropriate configuration of mental resources, which is a task set, and it takes time for people to initiate a different task set (Monsell, 2003). There are two main types of switch cost, *local switch cost* and *global switch cost*. An example of a task switching paradigm is shown in Fig. 2. This paradigm is composed of task switching trials and task repetition trials. The reaction delay and performance deterioration caused by switching between tasks is called *local switch cost* (Schneider & Logan, 2009). Combined with the paradigm, the performance (in terms of RT and error rate) of trial 4 (switched from trial 3) is usually better than trial 3 (switched from trial 2), and the improved performance is attributed to the lack of a local switch cost (Strobach et al., 2012). Additionally, in a paradigm composed entirely consecutive trials of Task A (or Task B), the responses are usually quicker and more accurate than when the same task is repeated in the task switching paradigm. In this situation, the



Fig. 2. An example of a task switching paradigm. Numerals from 1 to 10 denote the task trial number.

performance difference is caused by a *global switch cost* or *mixing cost* (Gamboz et al., 2009; Strobach et al., 2012).

When people change tasks, they first undergo a process of *task-set reconfiguration* which is analogous to mental 'gear shifting'. The time taken by this shifting can be shortened if sufficient time is given before switching stimulus onset, which is called the *preparation effect* (Monsell et al., 2003). The remaining switching cost is defined as *residual cost* which can last for more than 5 s (Kimberg et al., 2000; Sohn et al., 2000). Apart from preparation, three types of interference influence task switching performance: proactive task, task stimulus and response-based interference (Kiesel et al., 2010). The influence of proactive tasks is considered from switch-cost asymmetries and repetition cost. Switch-cost asymmetries refer to the surprising asymmetry in which it is harder to switch to a well-practiced (stronger) task from a less-practiced (weaker) task than *vice versa* (Monsell, 2003; Yeung & Monsell, 2003). This is because one has consciously inhibited the stronger task to perform the weaker task and switching back to the stronger task requires the inhibition to be lifted. In contrast, the stronger task is already dominant, so inhibition of the weaker task is not required (Allport et al., 1994). This asymmetrical switching cost exists especially in attentional switches between a perception task (usually requiring bottom-up attention) and a memory task (usually requiring top-down attention) (Hautekiet et al., 2023).

'Task stimulus' refers to the signal that announces the action of the switch. Switch costs are smaller when the stimuli are univalent (same stimulus for both tasks) than when they are bivalent (different stimuli for each task) because the stimulus is considered to contribute to the specific activation for a participant to retrieve the task and response (Allport et al., 1994). When the stimulus for each task is different, the stimulus can contribute to the retrieval. When the stimulus is the same for both tasks, the stimulus cannot contribute to specific retrieval. This univalent and bivalent situation also influences response-based interference. Switch costs are larger when the responses required for two tasks are different, as opposed to when the responses for both tasks are the same (Brass et al., 2003).

2.3. Driving-related task switching

Fig. 3 summarizes concepts of attention and task switching in the context of driving. When a driver engaged in a task that only involves automatic attention (e.g. driving on a highway without any traffic) receives a task switching stimulus (e.g. hearing a warning tone), a rapid response (e.g. braking) can be affected because the response is reflexive and requires little cognitive load. Although the driver can switch their attention rapidly, the appropriateness of the response might be high or low, depending on the predictability of the circumstances. However, when a driver engaged in a complex cognitive task that requires focussed attention (e.g. texting on a mobile phone) receives a task switching stimulus, the driver switches their attention to driving more slowly and with a higher likelihood of responding inappropriately. This is because the driver must first shift their attention from the previous task and assess the driving conditions before deciding on how to respond. This attentional switch involves a high cognitive load, incurs high switch costs and is associated with slower responses and a higher likelihood of erroneous responses.

Driving involves complex cognitive tasks, including speed monitoring, responding to road signs, planning routes, and anticipation of changed traffic conditions. This requires drivers to multitask. Concurrent multitasking without task switching is difficult to achieve (MacPherson, 2018; Madore & Wagner, 2019; Pashler, 1994), and can only be achieved when the responses to the tasks are automatic, solidified in LTM (Skaugset et al., 2016). Simultaneous performance of non-automatic tasks is not possible because performance of these tasks requires conscious, deliberate attention and they are limited by the capacity of WM (Schneider & Shiffrin, 1977). While safe driving depends on automated responses, drivers must continuously process information from the environment and vehicle and be able to respond to changed conditions. Such adaptive responses impose a cognitive load and involve WM. WM plays a crucial role in information processing during driving (Zhang et al., 2023). In the present paper, when discussing driver multitasking, such as distracted

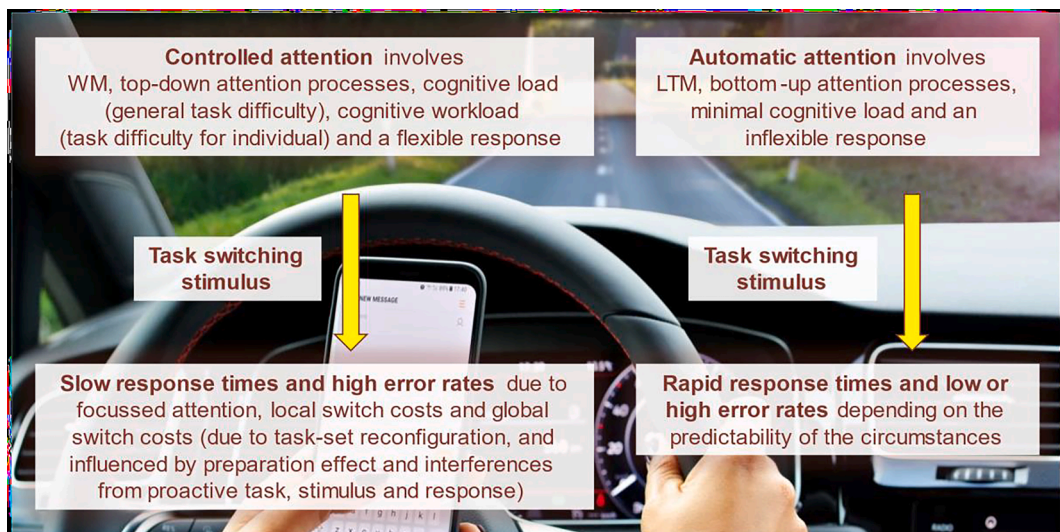


Fig. 3. Cognitive psychology concepts related to driving task switching and their relationships.

driving, the paper is referring to frequent switching between the driving task and secondary tasks.

In level 3 conditional automated driving, drivers are permitted to engage in NDRTs until they decide to take control of the vehicle or are required to do so due to an authority limitation or system failure. In such situations, drivers must switch their attention from the NDRT in which they are engaged in the task of driving. Several recent studies have examined driving performance in response to takeover requests and have identified factors that influence the quality and speed of takeover performance (Hungund & Kumar Pradhan, 2023; McDonald et al., 2019; Morales-Alvarez et al., 2020). However, such studies have given little attention to the cognitive underpinnings of takeover performance.

3. Review method

The primary focus of this review is to elucidate the cognitive processes that occur when a driver in a level 3 conditional automated vehicle is required to shift their attention from engaging in NDRTs to taking control of the moving vehicle. Since similar cognitive processes occur during distracted driving in level 0–2 automated vehicles, distracted manual driving is also reviewed.

This comprehensive narrative review delves into the relationship between driver cognitive load and cognitive tasks as secondary activities during manual driving and conditional automated driving (ranging from level 0 to level 3 automation). To compile relevant data, an extensive search was conducted through electronic databases, including Google Scholar, Scopus, PubMed, and ScienceDirect. Relevant keywords regarding cognitive load and task switching during driving were applied, including effect / influence of cognitive task / standardised task / driving distraction / multitasking / task switching / on workload / cognitive function / mental workload / cognitive load during driving / automated driving / driving. The selected data sources encompassed a variety of scholarly outputs, including peer-reviewed manuscripts, conference proceedings, book chapters, books, reports, and web pages, all available in English. While the review did not impose a restriction on the publication year, preference was given to studies published within the last 5 years to ensure a contemporary perspective on the subject.

This review investigates the relationship between cognitive tasks and driving task performance. Since it is technically challenging to quantify the effect of naturalistic tasks on driving performance precisely, most studies in this area have used standardised cognitive tasks instead. For standardised cognitive tasks, task and driver engagement levels can be aligned to a range by comparing the effects of different cognitive tasks. While most studies included in this review used standardised cognitive tasks, studies that used naturalistic tasks such as interactions with cell phones, were identified, compared, and discussed, enriching the overall analysis.

Papers were excluded from consideration by the present review if: (a) driving performance and cognitive task performance were not measured or shown; (b) the cognitive task was not a quantifiable variable or the cognitive task was not applied in direct association with the driving task; (c) the paper presented data that had previously been published by the same author; (d) the paper focused on participants with pre-existing impairments of cognitive function.

4. Cognitive load in driving

After a comprehensive database search and screening, 29 papers, consisting of 13 manual driving papers and 16 conditional automated driving papers were found to satisfy the selection criteria and were recorded in this review. Key observations arising from review of these papers are discussed in the following sections.

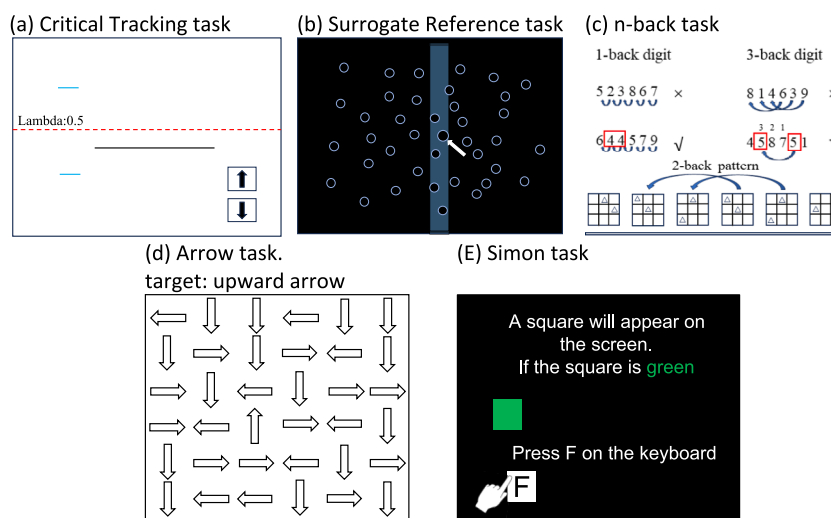


Fig. 4. Schematics showing concept of common cognitive tasks in driving performance research (a) Critical Tracking task, (b) Surrogate Reference task, (c) n-back task, drafted from the standard (ISO/TS 14198:2019); (d) Arrow task, drafted from (Hamish Jamson & Merat, 2005) and (e) Simon task, drafted from (Peng et al., 2023).

4.1. Types of driving tasks and cognitive tasks

Driving in a simulator is the most common and safest method for evaluating driving performance in response to task switching and changes in cognitive load (Blane, Falkmer, Lee, & Dukic Willstrand, 2018; Devlin, McGillivray, Charlton, Lowndes, & Etienne, 2012; Lew, Rosen, Thomander, & Poole, 2009; Vance, Fazeli, Ball, Slater, & Ross, 2014; Anderson, Rizzo, Shi, Uc, & Dawson, 2005). When investigating cognitive load, the driver is required to perform secondary tasks, which can be naturalistic or standardised. While naturalistic tasks resemble real-world situations, allowing for studying issues that occur in everyday life (Oviedo-Trespalacios et al., 2016), it is difficult to quantify cognitive load and participant engagement levels. Consequently, standardised cognitive tests of WM are preferred as they provide objective and quantifiable outcomes (von Janczewski et al., 2021). The International Organization for Standardization (ISO/TS 14198:2019) has recommended that the Critical Tracking task, Surrogate Reference task (SuRT) and n-back task be used as calibration tasks (secondary task) for the assessment of driver demand due to the use of in-vehicle systems. It also recommended the Lane Change Test (LCT) and Detection-Response Task (DRT) be used as the primary tasks. Other primary tasks include navigating, path following, manoeuvring, avoiding obstacles, and controlling speed (ISO, 2019).

LCT, as one of the primary tasks, is conducted on a real or simulated three-lane straight road with a series of lane changes defined by signs. In this test, a participant is instructed to follow the signs that appear at approximately regular intervals on both sides of the track to drive in the designated lanes. ISO (ISO/TS 26022:2010) recommends using the LCT in a dual-task scenario to measure human performance degradation on a primary driving-like task while a secondary task is being performed (ISO, 2010).

Cognitive tasks that are commonly used in driving research are shown in Fig. 4. In the Critical Tracking Task, the driver is required to prevent a horizontal bar presented on a computer screen from deviating from a predetermined position (Petzoldt et al., 2014) as shown in Fig. 4a, however, this task is infrequently used even though it is a standard task. In the SuRT, drivers are shown several circles of equal size alongside one larger target circle and are required to identify the target circle (Fig. 4b). The task difficulty varies based on the size difference between the target circle and the others, ranging from easy to challenging demand levels. The n-back task is a recall task that can be presented visually or audibly, where participants must indicate whether the same stimulus was presented to them on a previous occasion (Fig. 4c). The difficulty of this task can be manipulated by changing the value of 'n'. For instance, 1-back refers to the preceding presentation, while 3-back refers to the stimulus that was presented three presentations earlier. A review of the effect of n-back tasks on cognitive load in manual driving found a medium-to-large effect size on degraded driving performance (von Janczewski et al., 2021). While this confirms the n-back task as a useful tool for studying driver cognitive load, the present authors prioritised papers using the n-back task published since 2021, to avoid duplicating the content of Janczewski's review.

Additional psychological cognitive tasks have been used in driving research. The Arrow task requires participants to select an arrow in a 4×4 or 6×6 matrix to match the target (Hamish Jamson & Merat, 2005). In the Simon task, participants are required to respond to two different stimuli (from left or right) using two different buttons (left or right). They respond faster and more accurately if both the stimulus and response features are congruent (when the stimulus and the response are on the same side), and slower and less accurately when they are incongruent (when the stimulus is on the other side to the response) (Boudreau & Strybel, 2022).

Requiring a driver to perform a secondary task imposes additional cognitive and attentional demands. The effect of increased cognitive load on driving quality is assessed by comparing driving performance to a control condition where the individual is required to focus on driving in the absence of any secondary tasks. The metrics used to assess driving quality include steering wheel-based parameters such as the standard deviation of the steering angle, lateral position-based parameters such as the time spent out of the lane and the standard deviation of lateral position (SDLP) (Xu et al., 2024), and velocity-related parameters such as the standard deviation of longitudinal velocity. Similarly, the extent of cognitive load can be assessed by comparing performance on the secondary tasks (e.g. error rate, reaction time (RT), and response rate) in the absence of the driving task.

In the DRT, the driver is required to press a button when they detect a simple stimulus, which can be visual, tactile or auditory, and then their RT and response rate are recorded. Technically the DRT is a secondary task, as it is additional to the task of driving, and imposes a cognitive load; nonetheless, the load is considered to be minimal and it is generally used in combination with other secondary tasks to assess the impact of those tasks on cognitive load (Stojmenova et al., 2017; Stojmenova & Sodnik, 2018; van Winsum, 2018). For this reason, the DRT is often regarded as a primary task.

In summary, driving performance researchers use DRT and other driving-related tasks as the primary tasks, and cognitive tasks that are not related to driving to elevate drivers' cognitive load as the secondary task or NDRTs.

4.2. Cognitive load in manual driving

From SAE automation level 0 to level 2, driving is the primary task for drivers. This is referred to as manual driving, regardless of whether a manual or automatic transmission is involved. Drivers must monitor their environment and maintain safe driving, which demands ongoing cognitive involvement. Numerous studies have examined the deleterious effects of factors such as old age (Depestele et al., 2020), disease (Dobbs et al., 2005; Ranchet et al., 2012; Shimonaga et al., 2021), alcohol intoxication (Garrisson et al., 2021), and medications (Giorgetti et al., 2015).

For drivers with normal cognitive function, multitasking has a significant impact on road safety. This is because engaging in additional tasks divides the driver's attention and elevates their cognitive load, which in turn, increases the risk of errors (Regan et al., 2011). The slower reaction times and disrupted performance lead to a higher rate of accidents (Overton et al., 2015). While driving distraction can also occur in low cognitive load situations (e.g. mind wandering) (Martens & Brouwer, 2013), the present paper focusses on the high cognitive load imposed by secondary tasks.

Table 1 summarises 13 papers that evaluated driver cognitive load using different cognitive tasks. These papers primarily concern

driver distraction, multitasking and Advanced Driver-Assistance Systems (ADAS, applied in level 1 and level 2). Most cognitive tasks were presented audibly, visually, or in combination. Tasks such as the auditory version of the n-back task increased the driver's cognitive load without visual distraction. Studies using both visual and auditory tasks have found that visual distraction has a larger effect on driving performance than auditory distraction. For example, the auditory n-back task did not impair lateral out-of-lane performance whereas the visual n-back task led to significant impairment (Adachi et al., 2023). Similarly, using different types of visual and auditory cognitive tasks, it was found that the visual arrow task and SuRT decreased driving stability to a greater extent than the auditory n-back task (Ma et al., 2024; Yang, Ye, Easa, Feng, & Zheng, 2023). This can be explained using Multiple Resource Theory, which suggests that tasks relying on the same mental resources are more likely to interfere with each other (Wickens, 2008). Since driving relies heavily on visual processing, visual tasks like the arrow task and SuRT are more likely to disrupt driving performance compared to auditory tasks. This finding is further supported by research using naturalistic tasks. Wechsler and colleagues, for example, investigated driver multitasking with tasks such as typing, reasoning, and memory, and found that visual tasks had a more negative impact on driving than auditory tasks (Wechsler et al., 2018). Compared to other distraction sources, mobile phone use is becoming increasingly a concern. Researchers reviewed the impacts of mobile phone distraction on driving performance, finding that texting while driving, which requires both manual and visual attention, can cause visual, manual, cognitive or even auditory distraction, leading to serious traffic incidents (Oviedo-Trespalacios et al., 2016; Voinea et al., 2023).

In Table 1, some papers were primarily interested in driving performance and used cognitive tasks to increase drivers' cognitive load, while others used performance on cognitive tasks to evaluate the driver's cognitive load. Most papers found that driving performance was impaired when cognitive load was increased by having to perform secondary tasks. For instance, when the DRT was applied as an additional cognitive load, drivers' SDLP increased, indicating that the extra cognitive load reduced driving stability (Öztürk et al., 2023).

Furthermore, the SuRT caused drivers to spend more time out of the lane centre in both level 0 and level 2 manual driving, and conversely, drivers had a higher SuRT score when the cognitive load of the driving tasks was lower (Yang et al., 2023).

In manual driving, if a secondary task is introduced, the driver must divide their attention between the driving task and the secondary task, leading to a decline in performance on both tasks. When a 2-back task and DRT were applied in a car-following scenario, it was found that compared to a control condition where there was no extra cognitive task, driving stability decreased in terms of car position control and speed control (Zhai et al., 2020). When the 1-back and 2-back paradigms were compared, it was found that DRT performance and driving performance decreased more significantly in the 2-back condition, indicating that higher levels of cognitive load have a progressively negative impact on driving performance (Öztürk et al., 2023).

To summarise the main findings of Table 1, secondary tasks, especially visual ones, impair driving performance in manual driving. The degree of impairment increases as the task load increases, and performance on secondary tasks also suffers as attention is divided between the secondary task and driving. It is notable that most of these studies had relatively small numbers of participants (mostly 20–25) and the mean age of the participants was generally less than 30 years, so the generality of these findings to larger and more diverse populations remains to be determined.

4.3. Cognitive load in conditional automated driving

4.3.1. Measurements of driver performance

The primary parameters used to measure driver performance are vehicle-based metrics. In research involving level 3 conditional automated driving, participants are generally required to engage in NDRTs while the vehicle is under automatic control, to ascertain their effect on the speed and accuracy of the takeover response. Once the takeover request is launched, drivers are requested to resume manual driving. Vehicle-based driving performance enables objective comparisons of manual driving ability in the absence of an intervention (i.e. baseline manual driving) and following an intervention (e.g. after a takeover request).

In addition to driving performance, many studies have examined cognitive load following a takeover request. Table 2 summarises 16 recent papers on conditional automated driving that have used cognitive tasks. Eight of those papers assessed cognitive task performance with standardised tasks (Choi et al., 2020; Du et al., 2020; Melnicuk, Thompson, Jennings, & Birrell, 2021; Meteier et al., 2022; Radlmayr, Gold, Lorenz, Farid, & Bengler, 2014; Ucińska, Odachowska, Gąsiorek, & Kruszewski, 2021) and other tasks (Peng et al., 2023; Ucińska et al., 2021). The other 8 studies applied standardised cognitive tasks to differentiate drivers' cognitive load levels but did not evaluate performance on the cognitive task. The SuRT was used both as a visual NDRT in conditional automated driving and as a standard task to control participants' visual attention and cognitive engagement. Unlike studies with manual driving where n-back tasks were mostly presented aurally, in conditional automated driving, the n-back tasks were presented both aurally and visually.

Researchers choosing multiple types of cognitive tasks tend to differentiate cognitive demand types and observe their effect on driver takeover behaviours. Driver cognitive load variation during the takeover transition can also be followed by physiological parameters, such as the electrocardiogram (Meteier et al., 2022), heart rate variability (Du, Yang, & Zhou, 2020; Meteier et al., 2022; Chu, Wang, Deng, Lu, & Wu, 2020), galvanic skin responses (Du et al., 2020), respiration rate (Meteier et al., 2022) and electroencephalogram (Yang et al., 2023). Most papers recorded takeover driving performance using hands-on RT, lateral position change and steering behaviour. Eye-tracking is used to assess changes in the speed and direction of driver gaze (Du, Kim, et al., 2020; Feldhütter et al., 2017; Petermeijer et al., 2017). The most common method for subjective evaluation or self-report questionnaires is the NASA task load index. Other methods include the driving activity load index (Lu et al., 2021), visual analogue scale (Choi et al., 2020), and other questionnaires.

In summary, driver performance measurement includes vehicle-based driving performance, cognitive task performance, eye-tracking and gaze, physiological parameters and subjective questionnaires.

Table 1

Typical cognitive tasks used for the evaluation of driver cognitive load in manual driving.

Reference	Cognitive task type	Cognitive stimulus type	Measurement type	Effect of cognitive tasks	No./mean age of Participants	Main objective
(Hurts, 2011)	Spatial reasoning secondary task; acoustic secondary task	Visual; auditory	Driving performance; cognitive task performance	Both spatial/visual and acoustic/verbal tasks distracted driving. Spatial task was more distracting.	20/75.8	To ascertain how cognitive change in ageing affects driving
(Öztürk et al., 2023)	n-back with DRT	Auditory	Cognitive task performance; driving performance	Miss rate and response time of DRT Δ when n-back was applied, especially in 2-back. Without DRT, without n-back had the largest SDLP. With DRT, SDLP was similar.	20/22.0 17/66.0	The effect of cognitive load on DRT performance during day- and night-time driving
(Broadbent et al., 2023)	n-back	Auditory	2-back performance; Functional Near-Infrared Spectroscopy (fNIRS); subjective rating scale of mental effort; eye tracking; driving performance	Driving performance was not impacted by auditory n-back but changes in gaze behaviours, Δ physiological cost and Δ subjective report were observed.	21/23.9	Cognitive load, WM capacity and driving performance with fNIRS and eye tracking
(Adachi et al., 2023)	n-back	Auditory, visual	Reaction time and number of right answers of n-back; Driving performance; subjective achievement and effort evaluation.	Auditory n-back did not have a significant influence on driving performance but visual n-back ∇ driving performance in terms of lateral off-lane**.	18/37.6	Impact of visual and auditory cognitive load input on driving performance
(Huang et al., 2024)	n-back	Auditory	Electroencephalograph (EEG); Electrooculogram (EOG); Electrocardiograph (ECG); Electrodermal activity (EDA); driving performance	Compared 1-back to non-task: Δ subjective mental load (NASA TLX); Δ EEG α band power spectral density**; ∇ fixation distance*; Change in HR and HRV parameters *~***; Δ electrodermal activity (EDA) parameters*. Compared 2-back to 1-back: Δ subjective mental load; ∇ EEG α band power spectral density**; ∇ fixation distance*; Change in HR and HRV parameters * ~ ***; ∇ electrodermal activity (EDA) parameters* ∇ absolute steering*, ∇ SDD of acceleration*, Δ speed*, ∇ lateral position change*	24/24.5	Using multimodal to measure driving mental load during dual tasking
(Xue et al., 2023)	n-back	Auditory	Driving performance	From baseline to 1-back and 1-back to 2-back, ∇ driving stability in terms of steering wheel angle, steering wheel speed, longitudinal speed, headway distance and throttle angle.	24/24.0	Young novice drivers' cognitive distraction in a car-following scenario and machine learning classification
(Y. Yang et al., 2023)	Arrow task; n-back	Visual arrow task; auditory n-back	EEG; Driving performance; Eye movement	Compared to baseline, arrow task and 2-back Δ driver vigilance from EEG data, ∇ driving stability from longitudinal acceleration and steering wheel angle, ∇ attention time on driving, with larger influences by the arrow task.	32/24.0	Effect of visual and cognitive distraction on driver mental load
(Kaul & Jipp, 2023)	n-back	Auditory	Driving RT, Driving go/stop decision making; 1-back correct answer; pupil dilation	Compared to baseline, 1-back did not influence go/stop behaviour, but Δ RT, Δ cognitive load due to Δ pupil dilation	20/36.6	Influence of lead car and cognitive task on drivers' go/stop behaviour after yellow light-onset

(continued on next page)

Table 1 (continued)

Reference	Cognitive task type	Cognitive stimulus type	Measurement type	Effect of cognitive tasks	No./mean age of Participants	Main objective
(Ma et al., 2024)	SuRT; n-back; naturalistic tasks	Visual SuRT; Auditory n-back	Driving performance, DRT reaction time, eye glances	Compared to baseline, DRT RT was influenced significantly by 1-back. Glance off-road was Δ by SuRT. The standard deviation of speed was Δ by SuRT** and 1-back** Compared to 1-back, SuRT Δ SDLP**.	25/25.0	Impact of passenger's display on driver load and driving performance
(Ronca et al., 2024)	SuRT; other tasks	Visual	Subjective; gaze; EEG; driving performance	SuRT subjectively distracted drivers. It Δ speed standard deviation and EEG-based distraction index on city and highway roads.	25/26.0	An EEG-based assessment of distraction
(Mamo et al., 2024)	n-back	Auditory	Driving performance; n-back correct rate	Percentage of correct lane changing decreases with increased n-back level.	68/22.1	The impact of cognitive load on a (LCT) among male autistic individuals
(B. Yang et al., 2023)	SuRT	Visual	Subjective; driving performance; gaze; scores of SuRT	Compared to baseline, ∇ percent in road centre, Δ gaze time to side mirrors, ∇ gaze time to speed monitor, ∇ blink rate and percentage of eyelid closure were found with SuRT in manual driving and level 2 driving. Δ scores of SuRT with level 2 driving.	20/26.8	Influence of Level 2 automated driving on driver behaviour compared to manual driving
(Zhai et al., 2020)	n-back; DRT	Visual; auditory	Driving performance	Driving stability ∇ regarding speed control, lateral position, and time to collision compared to no cognitive task.	24/33.0	The effect of cognitive load on driving performance in different driving scenarios
(Brookhuis et al., 2009)	Peripheral Detection Task (PDT)	Visual	Heart rate and HRV; PDT performance; subjective	With the congestion assistant, PDT missed signals percentage and RT Δ when congestion started.	37/-	To evaluate the mental load and acceptance of driving with a congestion assistant

Abbreviations: Δ , increased; ∇ , decreased; ns, no significant difference; *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; baseline: driving without cognitive task.

4.3.2. Effects of cognitive tasks

The impact of cognitive tasks on driver takeover performance was detailed in Table 2, outlining two distinct categories. The first category examines the effect of cognitive load during automated driving on driver takeover performance. The second category examines how different types or levels of cognitive tasks during automated driving affect driver takeover performance. In the first category, nine papers compared the effect of cognitive tasks in conditional automated driving. Eight of these papers found that engaging in cognitive tasks during automated driving impaired driving performance, as indicated by slower takeover RT or response time, decreased driving stability, increased subjective demand evaluation, and changed physiological parameters. However, one study found that the takeover RT was longer when there was no cognitive task, although the difference was not statistically significant (Lu et al., 2021). This exception may be due to cognitive underload. Automation reduces operators' mental workload, and mental underload can be as detrimental as overload, because attentional capacity shrinks to accommodate reductions in mental workload, and the shrinkage is responsible for the performance decrement (McWilliams & Ward, 2021; Young & Stanton, 2002a, 2002b). Lu and colleagues discovered that under high driving task loads (such as manual driving and level 2 driving), drivers exhibited deteriorating performance (such as lane change or crash) with increasing cognitive task complexity. Conversely, under lower driving task loads (in level 3 assisted driving), driving performance showed less susceptibility to the cognitive task. These findings align with the idea that driving and cognitive task loads combine additively, leading to poorer driving performance.

In the second category, five studies investigated the impact of various levels of n-back tasks. Four of these studies concluded that as the difficulty of the n-back task increased, driving performance declined in terms of reaction time (RT) and driving stability. Elevated heart rate and changes in respiration rate correlate with driving performance suggesting that heightened cognitive load affects drivers' physiological performance (Melnicuk et al., 2021). In addition, it was observed that the n-back performance decreased when the difficulty of the n-back task increased (Melnicuk et al., 2021; Meteier et al., 2022). The five papers also compared different types of cognitive tasks. While differences in driver performance were observed with different types of cognitive tasks, no consistent finding

Table 2
Cognitive tasks used in conditional automated driving research.

References	Cognitive tasks	Driver performance measured	Effect of cognitive task on driver takeover performance		No./mean age of participants	Lead time (s)	Scenario and road condition	Main objective
			Compared to non-cognitive task driving	Comparison between different tasks				
(Petermeijer et al., 2017)	2-back	D, E, S	∇correct driving response rate, Δtakeover RT but <i>ns</i> ,		18/44.0	7	Braking and/or steering to avoid stationary cars.	Comparing static and dynamic vibrotactile seats.
(Du et al., 2020)	1- and 2-back	C, D, E		From 1-back to 2-back: Δtakeover RT***, ∇takeover readiness***; Δeyes-on RT***, interaction effect between cognitive load and traffic density**	80/22.8	4 and 7	Vehicle automation reached the performance limit.	Cognitive load, time budget and traffic density on driver takeover performance.
(Chai et al., 2024)	2-back	D, S		Comparing auditory 2-back to visual 2-back: Δtakeover time, Δlane change time, ∇Maximum resultant acceleration, Δmin TTC; Δoverall workload;	32/22.8	8	Steering to avoid traffic accidents, obstacles, work zones and broken-down truck	Effects of various auditory takeover requests considering visual and auditory NDRTs.
(Liang et al., 2024)	0-back, 1-back, and 2-back	D, S		From 0-back to 2-back: ∇ perceived driver capability especially in high traffic density	57/38.51	7	Steering or brake to avoid traffic accidents in 3 levels of traffic density conditions	Examining the role of driver perception in takeover time.
(Chu, Wang, Deng, Lu, & Wu, 2020)	2-back	D	∇correct response rate to the stimulus but <i>ns</i>		3/NA	7.2	Lane changing to avoid stationary vehicles.	Vibration pattern on the seat and takeover response.
(Lu et al., 2021)	1- and 2-back	D, S	∇takeover RT, Δabnormal performance in high demand scenarios, ∇abnormal performance in low demand scenario	From 1-back to 2-back: ~takeover RT, abnormal performance in 3 high demand scenarios, ~abnormal performance in low demand scenario	19/31.6	2 and 5	Decelerating or lane changing to avoid a crash with a lead car or stationary vehicle.	Driver takeover performance in varying levels of automation.
(Du et al., 2020)	1- and 2-back	D, P, E		From 1-back to 2-back: ∇HRV*, ∇horizontal gaze dispersion***, ∇Eyes-on-road time***, Δblink numbers after takeover;	102/22.9	4 and 7	Lane changing for 8 different events; no traffic.	Psychophysiological responses to takeover request.
(Meteier et al., 2022)	1- and 3-back	C, P, S	Δsubjective mental demand***; Δtonic level of electrodermal activity in visual n-back tasks but <i>ns</i> , the difference between visual and auditory tasks***, ΔHR in 3-back***, Δchange in respiratory rate***	From 1-back to 3-back: ∇n-back performance*** ~tonic level of electrodermal activity, ΔHR***, Δchange in respiratory rate***; visual 1-back more demanding than auditory 1-back**, but visual 3-back	80/23.9		Takeover for 5 different events.	Classification and regression for assessing load using physiological indicators.

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Table 2 (continued)

References	Cognitive tasks	Driver performance measured	Effect of cognitive task on driver takeover performance		No./mean age of participants	Lead time (s)	Scenario and road condition	Main objective
			Compared to non-cognitive task driving	Comparison between different tasks				
(Melnick et al., 2021)	0-, 1-, and 2-back	C, D, P, S		not much demanding than auditory 3-back* From 0- to 2-back: n-back Δ error rate, Δ missing rate; Δ DAI scores; Δ mean speed, SDLP and stabilisation time, ∇ mean range and mean lateral position; mean heart rate lowest at 2-back, HRV most affected by 1-back in the urban scenario	38/33.5	10		Driver cognitive load and stabilisation time on motorway and urban.
(Chen et al., 2021)	SuRT	D	Δ disengagement time*, Δ takeover time*, ∇ minimum TTC but <i>ns</i> , Δ resultant acceleration*** in experienced group but <i>ns</i> in novice group, Δ maximum lateral acceleration*** in 5 s time budget but <i>ns</i> in 7 s		24 + 24/26.8 + 35.8	5 and 7	Evasive manoeuvres for the sudden appearance of broken-down vehicles.	Takeover ability of novice and experienced drivers.
(Feldhütter et al., 2017)	SuRT	D, E	∇ TTC, Δ takeover time**; ∇ number of gazes towards the driving scene**		30/24.2	6	Avoiding suddenly appearing stationary vehicles.	Effect of duration of automation and gaze behaviour.
(Ucińska et al., 2021)	1-and 2-back, SuRT, Reaction time test	C (Reaction time test and SuRT), D	SuRT: Δ RT in young, middle-aged, and elder people**; Δ response time in all driving experience groups**.	<i>Only SuRT and a naturalistic task were used in conditional automated driving.</i> SuRT compared to fog condition: ∇ RT in young and middle age groups, Δ RT in the elder group; ∇ RT in brief and moderate years of driving group, Δ RT in long-term drivers***	96/35.1		Lane changing or braking for traffic incident site.	Age and experience in driving and psychomotor skills in automation.
(Peng et al., 2023)	SuRT, 2-back, Simon task, Visuospatial WM task	C (Simon task, visuospatial task), D, S	SuRT: Δ standard deviation of steering angle*; n-back: Δ subjective effort*** and fatigue***, ∇ standard deviation of steering angle <i>ns</i> ; No NDRT: Δ subjective bored*** and sleepiness***	Compared to SuRT, n-back: ∇ standard deviation of steering angle <i>ns</i> , ∇ SDLP <i>n</i>	24/67.6	4 and 8	Lane changing to avoid stationary vehicles, one vehicle on the target lane.	Elder drivers' cognitive ability and takeover performance.
(Liang et al., 2022)	SuRT, PDT	D, S		Compared to PDT, SuRT ∇ appropriate manoeuvring percentage	43/21.2	7	Driving manoeuvre to avoid collision; nighttime.	Effect of NDRTs on takeover performance during nighttime.
(Radlmayr et al., 2014)	SuRT, 2-back, DRT	C (DRT), D, S		DRT RT significant difference between SuRT and n-back**	48/33.5	7	Lane changing for obstacle avoidance; different traffic	Traffic situation and NDRT affecting takeover quality.

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Table 2 (continued)

References	Cognitive tasks	Driver performance measured	Effect of cognitive task on driver takeover performance		No./mean age of participants	Lead time (s)	Scenario and road condition	Main objective
			Compared to non-cognitive task driving	Comparison between different tasks				
(Choi et al., 2020)	SuRT, 2-back, 1-back	C, D, S	N-back group: ∇response time <i>n</i> , Δtime from steering initiation to lane crossing*, Δsteering angle variation during and after lane crossing but <i>ns</i> . SuRT group: ~response time <i>n</i> , Δtime from steering initiation to lane crossing <i>ns</i> , ~steering angle variance from steering initiation to lane crossing <i>ns</i> , Δsteering angle variance after lane crossing <i>ns</i>	Subjective rating: ∇accuracy* and Δsubjective difficulty score*** for both n-back and SuRT as Δtask difficulty <i>N-back tasks and SuRT were conducted by different groups of people, and no comparison was available.</i>	28 + 32/38.8 + 36.5	6	conditions on one of three lanes. Car following and lane changing for broken-down car avoidance.	Cognitive load and visual load on driving takeover.
(Lin et al., 2021)	SuRT, Arrow task	D		No difference between different levels of the arrow task. No difference between arrow tasks and SuRT. Compared to naturalistic tasks, cognitive tasks Δhands-on time, but <i>ns</i>	48 + 49 + 21/31.2 + 31.0 + 33.1	5 and 7	Broken-down vehicle avoidance; high and low traffic density levels.	Effects of NDRTs on readiness to take over the vehicle.

I. Abbreviations: RT, reaction time; SDLP standard deviation of lateral position; TTC, time to collision; *ns*, no significant difference; *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$;

II. Driver performance measurement: C, cognitive task performance; P, physiological measurements; D, (vehicle-based) driving performance; E, eye-tracking measurements; S, Subjective evaluation.

III. Effects: Δ increased ∇ decreased ~ similar to.

emerged, due to a small number of papers. Visual tasks negatively affected driver RT more than auditory tasks (Choi et al., 2020; Deng et al., 2019; Ucińska et al., 2021), presumably because driving is primarily a visual task, and switch cost results in longer RTs with visual tasks (Strobach et al., 2012) according to Multiple Resource Theory (Wickens, 2008).

To summarise the preceding findings, most researchers found that engaging in NDRTs leads to an increased cognitive load, which impairs a drivers' ability to control the vehicle during the takeover response. When the difficulty level of the NDRTs increased, driving takeover was more greatly affected. Cognitive underload during the NDRTs also impaired the takeover response. It was also found that visual NDRTs negatively influence takeover performance more than auditory NDRTs. This difference arises because driving is primarily informed by visual information, so switching attention from a visual NDRT to driving involves a larger switch cost. This concept is discussed in the following section.

5. Task switching in driving

In task switching experiments, a task typically involves performing a mental operation or action in response to a stimulus (Kiesel et al., 2010). Task switching refers to the flexible transition from one task to another (Parasuraman & Caggiano, 2002), where attention shifts from one task to another either voluntarily or involuntarily (Mochi & Madjar, 2018). In the context of driving, task switching refers to a driver shifting their attention between driving and a distracting task, or *vice versa*. Task switching during driving is common in daily life, such as reading a message on a phone or using an in-vehicle navigation function while driving. In conditional automated driving, drivers can focus on NDRTs and switch back to driving when requested. Since both conditions are similar to task switching scenarios used in psychological research (Shi & Bengler, 2022b), task switching theories can help us to understand why secondary tasks impair driving performance (Hecht et al., 2020; Shi & Bengler, 2022a).

5.1. Similarity of cognitive task switching and driving task switching

Task switching during driving is illustrated diagrammatically in Fig. 5 and Fig. 6. Fig. 5 relates to manual driving where the driver is required to simultaneously perform a secondary task, whereas Fig. 6 relates to automated driving where the driver is required to switch from an NDRT to taking over manual control of the vehicle. In manual driving, driving while performing a secondary task is a form of multitasking. Multitasking research includes research on dual-tasking performance, and task switching performance (Pashler, 2000; Strobach et al., 2018). When a secondary task is introduced (Fig. 5), attention may be distributed in one of two ways: either both tasks are performed simultaneously (block 1) or there is alternating switching between tasks (block 2). According to cognitive and attention research, dual tasks can be performed simultaneously when at least one of the tasks is automatic. It is very difficult to perform dual tasks, when both are non-automatic due to the limitations of WM (Skaugset et al., 2016). Driving is an automatic process for experienced drivers, particularly in straightforward conditions like highway driving. However, once the driving task becomes complex (Siéoff, 2001), for example, increased traffic density, drivers have to switch their attention to driving. In this case, dual tasks can only be performed when the driving task remains easy enough that the driver can continue to drive while performing the secondary task for a short time. In most instances, drivers rapidly switch their attention between the driving and secondary tasks to continue driving and performing the secondary task.

Fig. 6 relates to task switching during conditional automated driving. In Fig. 6, the NDRT is similar to task A in Fig. 2, and the resumption of manual driving can be considered task B. The switch can be initiated by the subjective intention of the driver or at the request of the automation system. Most researchers have investigated the latter situation, as shown in Table 2. The takeover request launched by the vehicle system is like the stimulus that calls for a task switch in the task switching paradigm, while the switch back to an NDRT from manual driving may not involve an overt stimulus. Road safety concerns are more likely to occur during the transition from automated to manual driving.

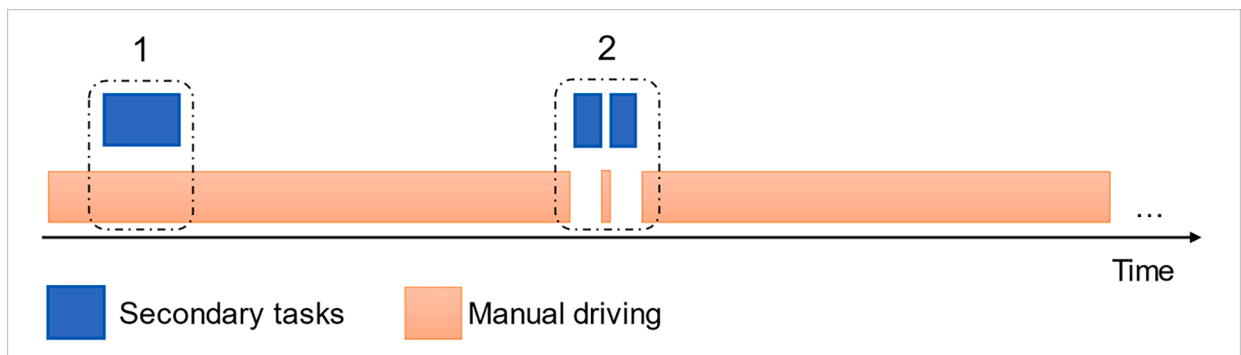


Fig. 5. A schematic of possible performance patterns during driving with a secondary task, draft based on combined concepts from (Shi & Bengler, 2022a), (Skaugset et al., 2016) and (Strobach et al., 2018).

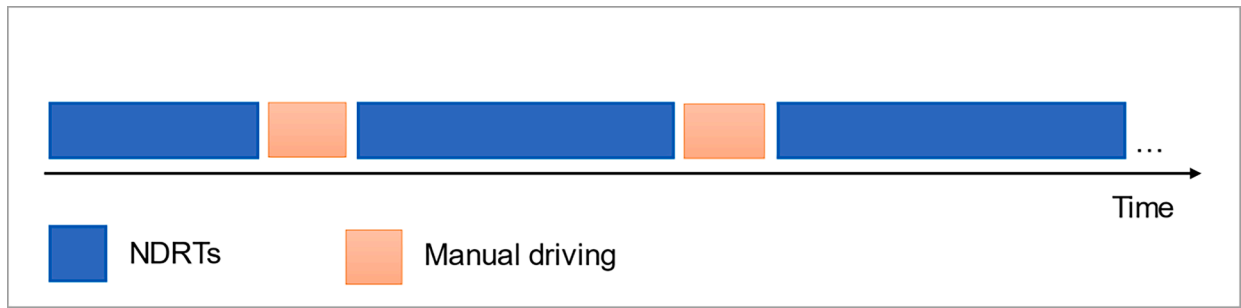


Fig. 6. A schematic of task switching during conditional automated driving, modified from (Shi & Bengler, 2022a).

5.2. Effects of switch cost on driving performance

Researchers have found that a switch cost exists if participants are involved in the task switching paradigm; switching costs lead to a temporary impairment of performance on both tasks. Local switch cost refers to the delayed reaction time and decreased accuracy rate that is immediately seen when the task is switched (Strobach et al., 2012). Consequently, when drivers switch from performing a cognitive task to driving, their reaction time and driving stability can be affected. Indeed, as shown in Tables 1 and 2, most studies have found that compared to not engaging in any cognitive task, switching from a cognitive task to a driving task resulted in increased reaction time and decreased driving stability in both manual driving and conditional automated driving. In theory, cognitive task performance after the switch back from driving should also be affected. However, this remains a topic for future research.

Local switch cost relates to the instant influence of task switching during driving, whereas global switch cost, or mixing cost, relates to the concept that the performance of each task (even a repeated task in mixed blocks) is not as good as when only one task is performed (Gamboz et al., 2009; Strobach et al., 2012). This concept predicts that driving performance will be impaired for a protracted period after task switching. This prediction is supported by evidence that driving performance is better in pure manual driving than after takeover in conditional automated driving (Lu et al., 2021; Radlmayr et al., 2014).

Since switch cost impairs driving after task switching, the duration of the switch cost is a vital issue for road safety. While most studies have demonstrated that it takes 20–40 s to normalise driving control following a takeover request (Merat, Jamson, Lai, Daly, & Carsten, 2014), estimates of the duration vary widely. At one extreme, it has been reported that driver SDLP and steering angle recover to normal levels 5 s after drivers resume control following a takeover request (Naujoks et al., 2019). A 5 s recovery time is remarkably short and is probably due to a minimal residual cost with adequate preparation, as shown by task switching research implying that the residual cost may last for 5 s in predictable circumstances (Kimberg et al., 2000; Sohn et al., 2000). At the other extreme, it has been reported that novice drivers can still be impaired 5 min after the takeover request and the drivers are unaware that their driving performance has degraded (Zhang et al., 2023). In that study, drivers were instructed to fully engage in the NDRT and ignore the driving conditions until requested to takeover. This naturalistic scenario resembles what will occur in level 3 automation and indicates that high levels of focused attention can greatly increase the duration of the switching cost.

Another factor influencing the duration of the switch cost is task difficulty of the takeover scenario. When different road conditions were used (in terms of accident ahead and traffic density) it was found that the easiest road condition allowed drivers to stabilize the vehicle in 12 s compared to 23 s and 29 s for the accident condition and accident with congestion condition, respectively (Kim et al., 2021). In addition to driving parameters, physiological measures can indicate the duration of physical recovery from driving task switching. Researchers measured heart rate and heart rate variability during and after driver takeover, and their findings indicated that heart rate parameters take 20–30 s to stabilize following a low-demand planned takeover and 60 s to plateau (Melnicuk et al., 2021).

5.3. Factors influencing task switching during driving

Factors known to influence task switching in cognitive psychology are expected to influence task switching in driving scenarios. Task switching knowledge can help us identify factors likely to adversely affect driving performance so that road safety can be improved.

5.3.1. Preparation effect

The primary factor that influences task switch performance is the preparation effect, which relates to the time needed to mentally prepare a response to an anticipated stimulus. If insufficient preparation time is available, the response to the stimulus will be delayed. Prolonging the time interval before stimulus onset can reduce switch costs (Kiesel et al., 2010). Therefore, time budget, or lead time is considered to directly contribute to driving switch cost. Research consistently highlights the impact of time budget on switch costs in driving scenarios. When switching from NDRTs to driving, those drivers allotted a longer lead time, ranging from 6 to 10 s, exhibit greater stability in their driving performance compared to those with a shorter time budget of less than 5 s (Chen, Lu, Lin, Zhai, & Tan, 2021; Du, Kim, et al., 2020; Du, Yang, & Zhou, 2020; Lin, Lyu, Zhang, & Ma, 2021). The longer time budget allows drivers to benefit from the preparation effect to switch back to driving more effectively. These findings highlight the importance of allowing enough time for task preparation in conditional automated driving scenarios, leading to a safer and more controlled driving experience.

The quality of preparation time is another factor. To effectively reduce switch costs, switch-specific preparation time is important, rather than temporal preparation (Kiesel et al., 2010; Rogers & Monsell, 1995). A long lead time is not advantageous if attention is being directed to another task during that time. By applying this point of view to task switching in automated driving, information about upcoming takeover tasks should be introduced to drivers before the occurrence of switching to allow for effective preparation. For example, when different levels of explanations were given regarding upcoming takeover scenarios, those participants with more detailed explanations displayed a significantly higher rate of successful interventions (Zhou et al., 2021). Therefore, in driving task switching, clear and targeted instructions help drivers to switch more effectively.

5.3.2. Interferences in task switching

As mentioned earlier, interferences in task switching come from proactive task interference, stimulus-based interference, and response-based interference. The following subsections discuss the interferences occurring in driving task switching.

5.3.2.1. Proactive task interference. Interference from proactive tasks comes from switch-cost asymmetries (Kiesel et al., 2010). In task switching, one unconsciously inhibits the skilled task in order to perform the unfamiliar task. In automated driving the type of NDRT can affect driving performance. After performing a NDRT that requires their full attention, the driver may demonstrate an inability to properly control the vehicle after switching. This type of impairment has been demonstrated using different levels of n-back tasks. As the n level increased, drivers' reaction time and driving stability after the takeover worsened (Du, Kim, et al., 2020; Du, Yang, & Zhou, 2020; Melnicuk, Thompson, Jennings, & Birrell, 2021; Meteier et al., 2022).

5.3.2.2. Stimulus-based interference. In task switching, stimuli are considered to trigger retrieval of the task and provoke a larger switch time, with univalent stimuli involving a smaller switch cost than bivalent stimuli (Waszak et al., 2003). In automated driving the stimuli are neither univalent nor bivalent, because a stimulus cues the switch from an NDRT yet the switch from driving to an NDRT is generally left up to the discretion of the driver. Nonetheless, the type of stimulus may influence the speed of switching from NDRT to driving. A study of various warnings, including A-pillar LED light, earcon, speech messages, and haptic seat feedback (Hong & Yang, 2022) found that while strong warnings such as the haptic seat and the A-pillar LED light marginally reduced RT, they concurrently elevated stress levels and compromised overall driving stability. While an enhanced RT is beneficial, the potential increase in stress levels and reduced stability denotes the need for a balanced approach when designing human-vehicle interaction systems. Striking this balance becomes vital in ensuring efficient task performance and the well-being and safety of the driver.

5.3.2.3. Response-based interference. An aspect of response-based interference is the response-repetition effect. This effect relates to the fact that repetition of a response improves performance of the repeated task, but at the cost of the switching task (Kiesel et al., 2010). Thus, repetitive responding to one task adversely influences the performance of the other task. This effect is similar to engagement in NDRTs during automated driving, because repetition of the NDRT task equates to spending a longer time engaging in this task. Indeed, two studies have demonstrated that engaging in an NDRT for 20 min worsens takeover performance and vigilance level to a greater extent than engaging in the same NDRT for 5 min (Feldhütter, Gold, Schneider, & Bengler, 2017; Zhang, Fard, Xu, Davy, & Robinson, 2023).

5.3.3. Other factors

Apart from factors correlated with task switching, other factors are known to influence the cognitive load either by influencing the driver or the driving environment. These factors are considered below.

5.3.3.1. Driver-related factors. Driver Age. Although there are substantial individual differences in WM capacity (Kotlyusov et al., 2023; Robison & Unsworth, 2017), there is a trend towards slower RTs and poorer WM as we grow older (Gajewski et al., 2018). Studies have reported poorer secondary task performance and worse driving performance among elder people when compared to younger people (Öztürk et al., 2023; Peng et al., 2022; Ucińska et al., 2021). While tasks involved the same cognitive load, the elder group may have experienced a higher cognitive workload, resulting in a deterioration in performance. Brain imaging studies have shown that increased age is associated with a strong increase in global switch costs whereas increases in local switch costs are more modest (Eich et al., 2023). This difference suggests that in conditional automated driving, elders may switch successfully from NDRTs to driving, yet their driving performance after switching may take longer to recover to control levels than younger people.

Driver Experience. Drivers acquire attentional skills and decision-making skills through experience (Drummond, 1989). Through practice, driving skills can be stored in LTM, which enables these skills to be employed automatically. Consequently, inexperienced drivers utilise more WM when driving and are more prone to cognitive overload. As a result, novice drivers experience a higher accident rate. For instance, in the State of Victoria, Australia, probationary drivers have a significantly higher risk of casualty crashes in both daytime and late night (VicRoads, 2023b). Similar results are observed for level 3 automated driving. Novice drivers have a similar takeover response time to experienced drivers, yet they have poorer takeover stability and weak adaptability in terms of lateral control in different urgency scenarios (Chen et al., 2021). In support of this finding, researchers reported that following a takeover response, inexperienced drivers spent significantly more time out of the lane and exhibited more erratic lateral control of the vehicle. Interestingly, years of driving experience were a less significant factor than the number of kilometres driven, with a maximum benefit gained from 10,000 km of driving per year and 100,000 km of driving in total (Zhang et al., 2023). These findings suggest that considerable driving experience is required to fully encode driving skills into LTM.

5.3.3.2. Environmental factors. Visibility. Visibility on the road, affected by weather and the time of day, significantly impacts driver cognitive load. Reduced visibility, especially during nighttime or in fog, demands heightened cognitive efforts for situational awareness. Research indicates that participants exhibit poorer driving stability in low-visibility conditions, underscoring the challenges drivers face during nighttime travel (Öztürk et al., 2023).

Road conditions. Road conditions play a pivotal role in shaping driver task switching performance, particularly regarding traffic density. Road conditions also include changes in road curvature, road configuration, changes in road surface conditions and so on. Recent research has focussed on traffic density. Heavy traffic requires more situational awareness and a greater number of decisions, which significantly increases the cognitive load for drivers, impacting their ability to efficiently switch between tasks (Du, Kim, et al., 2020; Du, Yang, & Zhou, 2020).

6. Discussion

6.1. Should distracted tasks be permitted in level 3 automated driving?

The present review has shown that the process of switching attention from one task to another may appear to be quick from an observer's standpoint, yet a complete switch of attention, as indicated by the normalisation of objective performance measures, can require between 20 s and 5 min. In the context of level 3 automated driving, it means that a driver will drive worse than normally (in some cases dangerously out-of-lane) for up to 5 min after resuming control of their vehicle following a takeover request.

For task switching and multitasking in manual driving, extensive research and discussions have highlighted the deterioration of driving performance and the increased rate of road accidents caused by distracting tasks (Llerena et al., 2015; Overton et al., 2015). Society has acknowledged the dangers of distracted driving, prompting legislation in most countries that limits the types of secondary tasks drivers can engage in. For example, in the State of Victoria, Australia, learners and probationary drivers are prohibited from using any mobile device, including mobile phones, while driving. In contrast, fully licensed drivers are permitted to make or receive phone calls and use navigation systems, provided they are not handheld (VicRoads, 2023a). This difference is due to the recognition that young drivers, who are often less experienced, are over-represented in serious road crashes. Also, the National Highway Traffic Safety Administration (NHTSA) issued driver distraction guidelines for in-vehicle electronic devices (NHTSA, 2012). This document advises against in-vehicle electronic systems that allow drivers to perform some activities when the vehicle is moving, especially visual-manual demanded activities. However, there appears to be a lack of discussion regarding restrictions on NDRTs in the forthcoming era of conditional automated driving.

It needs to be acknowledged that most current studies of conditional automated driving rely on data obtained in driving simulators, primarily due to regulatory constraints and safety concerns. How these findings will translate to real-world driving scenarios requires further investigation. Having said that, the driving simulator data that are currently available consistently demonstrate that task switching impairs driving performance, and there is no reason to believe that this will change in the real world.

The SAE standard for on-road motor automation systems (SAE, 2021) describes drivers in level 3 as *dynamic driving task (DDT) fallback-ready users*. This standard denotes that as DDT fallback-ready users in level 3 automation, drivers do not need to monitor the vehicle, yet they are expected to be prepared to either resume DDT performance or ensure minimal risk conditions. This means that drivers are responsible for actively switching their attention from other tasks to driving. The data in the present review show that drivers consistently perform worse when engaged in task switching than when they are not. This decline in performance can be mitigated to some extent but not eliminated. Additionally, different types of NDRTs have varying levels of negative effects on driver takeover performance. Hence, the concern persists regarding whether all NDRTs are safe enough for all drivers, particularly for inexperienced ones. In this context, evaluating the cognitive load of naturalistic demands becomes crucial. While cognitive psychology research has classified the levels of cognitive demand for daily tasks (Romero-Ayuso et al., 2021), the assessment of in-vehicle activities remains a topic of discussion.

In summary, these concerns raise questions for vehicle manufacturers, governments, and road users alike: What type of human-vehicle interaction design can effectively mitigate the risks associated with driver task switching, particularly in higher levels of automation? Are novice drivers or elderly drivers capable of effectively switching their attention back to driving? What types of NDRTs fall within a safe range for use on the road by drivers?

6.2. Suggestions to improve road safety with higher-level automation

The present authors recommend that before higher levels of vehicle automation are permitted on suburban roads, protections ought to be put in place to reduce the safety concerns raised by the inevitable task switching during driving. The following paragraphs outline potential ways to reduce the risks associated with driver task switching.

For vehicle Automated Driving Systems (ADS) design. Driver preparation can reduce switch costs, and a longer lead time free of cognitive tasks can improve driver takeover stability. With current research that is mostly based on young drivers, once the driver ceases engaging in an NDRT, they require 6 s to refocus their attention on the road conditions before taking control of the vehicle. Ideally, ADS should incorporate driver preparation time prior to the takeover. This suggestion has been addressed by the United Nations Economic Commission for Europe (UNECE) in UN Relation No. 157. This regulation requires a minimum of 10 s to be provided to transition to manual driving for an automated lane-keeping system (UNECE, 2021).

Furthermore, to assist takeover in level 3 automation, drivers will need to be repeatedly exposed to takeover requests under safe conditions, until they develop automatic responses to takeover requests that are stored in their LTM. Such training sessions should be a

feature of the ADS. Since there is individual variability in the speed and accuracy of takeover responses, the ADS should be able to build a predictive profile of the driver to anticipate the time required by the driver to return to a safe level of driving performance after a takeover. It follows from this that the ADS will need to be able to recognise when a different driver is behind the wheel. Finally, the ADS will require backup strategies that can be quickly implemented in the case of unsuccessful takeovers.

For regulations and laws. Regulatory authorities will need to prescribe which types and durations of NDRTs are acceptable for drivers to engage in during level 3 conditional automated driving. The permissibility of probationary licensed drivers using level 3 automation will require resolution before conditional automated driving is legally implemented. Authorities ought to consider what driving conditions are permissible for drivers to engage in NDRTs. Automated driving on arterial roads during the daytime in good weather may be acceptable, but do local roads at night, in rain, carry too much risk?

For drivers. Drivers should first acquire manual driving abilities before using level 3 conditional automated driving functions. It is recommended that drivers should first gain extensive manual driving experience to reduce the cognitive load of basic driving tasks. Additionally, drivers should refer to instructions to familiarize themselves with vehicle controls before using level 3 automation. Drivers need to be educated about task switching and the associated decline in driving performance, particularly as drivers report no awareness that their driving performance has declined after a takeover (Zhang et al., 2023). Another suggestion is that drivers will need to use the manual controls regularly, as there is a risk of diminishing automaticity due to infrequency of use. This is happening in the aviation industry where automation reduces the risk of human errors most of the time as the shown in the safety record, but automation has the potential to gradually diminish cognitive skills, which only becomes apparent when the pilot is required to respond exceptional circumstances (Oliver et al., 2017).

For further research. As driving task switching research is mostly based on driving simulators, on-road tests will be required to validate the results from simulation studies. In addition:

- I. Further studies are needed to identify which NDRTs are compatible with rapid and accurate takeovers, and which NDRTs most greatly compromise driving performance after a takeover. The duration of the NDRT also needs to be examined, as most studies to date have examined short durations of a few minutes, whereas the activity of watching a movie could potentially last more than an hour in real life. Engagement in these NDRTs should involve a maximal level of focussed attention since this is expected in real-life scenarios.
- II. The available data are largely based on healthy young university students, making it likely that these data are not representative of most drivers. Future studies of task switching in conditional automated driving should compare driver factors such as driving experience, driver age, driver gender or the influence of common health disorders (e.g. obstructive sleep apnoea, insomnia, obesity, subclinical depression, prescription medications). Notably, the population spread of performance after takeover needs to be understood, as the population mean is an unreliable indicator. Road accidents are more likely to be caused by those individuals who are outliers (e.g. 2 standard deviations above the mean) in terms of the duration required to return to normal driving performance after a takeover.
- III. While intervention and self-interruption have been found to reduce takeover RT, repetitive and unnecessary task switching can increase the cognitive load with minimal benefits (Y. Wu et al., 2019). From a road safety perspective, the optimal frequency of interruption from NDRTs remains to be determined. Insights into this could be gained from research on Driver Monitoring Systems (Braunagel et al., 2017; Guettas et al., 2020; Yu et al., 2021). DMS have been increasingly studied in recent years to detect and classify driver states—such as inattention, posture changes, fatigue, stress, and emotion—using physiological, behavioural, and vehicle-based approaches (Guettas et al., 2020; Hecht et al., 2019). By continuously monitoring driver states, DMS can help trigger timely interruptions when drivers are at risk of losing control, ensuring they successfully switch back to the driving task.

6.3. Limitations

The primary limitation of the present review is that it considers standardised tasks which would not occur during normal driving, instead of reviewing the effects of the types of distractions that would normally occur during driving. As we have noted, standardised tasks are widely used by researchers because they reliably and completely distract attention and can be readily replicated between studies. In contrast, real world distractions are inherently variable in duration and in the amount of attention they require, which makes them difficult to quantify in a laboratory setting. Nonetheless, there is a need for more research that uses distraction paradigms that are close to real-life situations. Another limitation of the scope of the present review is that, due to space limitations, we only considered studies with young and healthy participants. Research has been published on task switching in drivers with cognitive impairment due to factors such as degenerative disease (Ahmadnezhad et al., 2022; Ahmadnezhad et al., 2023), alcohol and drug intoxication (Wiedemann et al., 2018), or medication use, and these studies consistently show that these factors greatly lengthen the time taken to switch attention.

7. Conclusion

This review explained attention and cognitive processes involved in driving and applied task switching concepts to manual driving and level 3 conditional automated driving. Data consistently show that cognitive overload and switch costs lead to a significant delay in resuming control of the vehicle and driving safely. In manual driving, when drivers undertake distracting tasks, their performance for both driving and the cognitive task can be impaired. In conditional automated driving, switching between NDRTs and driving not

only slows reaction time, but it also impairs driving stability afterwards. This impairment can last for up to 5 min. While switch costs can be reduced, they cannot be eliminated. In order to minimise safety risks from level 3 automated driving, safeguards need to be put in place by vehicle manufacturers, legislators and drivers. There is also an urgent need for research to address unanswered questions.

CRedit authorship contribution statement

Jinhui Xu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Mohammad Fard:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Neng Zhang:** Writing – review & editing, Conceptualization. **John L. Davy:** Writing – review & editing, Supervision, Funding acquisition. **Stephen R. Robinson:** Writing – review & editing, Writing – original draft, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- Adachi, H., Muramatsu, Y., Sakamoto, N., & Ohya, H. (2023). The impact on driving performance from graded cognitive load with visuo-spatial and phonological processing of visual and auditory input 27th International Technical Conference on the Enhanced Safety of Vehicles (ESV) National Highway Traffic Safety Administration, Yokohama, Japan. <https://www.nrd.nhtsa.dot.gov/departments/esv/27th/>.
- Ahmadnezhad, P., Burns, J., Akinwuntan, A., & Devos, H. (2022). Cognitive Workload and Emergency Response during Partially Automated Driving in Preclinical Alzheimer's Disease. *Archives of Physical Medicine and Rehabilitation*, 103(12), e55.
- Ahmadnezhad, P., Burns, J. M., Akinwuntan, A. E., Ranchet, M., Kondyli, A., Mahnken, J. D., & Devos, H. (2023). Driving Automation for Older Adults with Preclinical Alzheimer's Disease. *Gerontology*, 69(11), 1307–1314. <https://doi.org/10.1159/000531263>
- Allport, D. A., Styles, E. A., & Hsieh, S. (1994). Shifting intentional set: Exploring the dynamic control of tasks. In *Attention and performance 15: Conscious and nonconscious information processing*. (pp. 421–452). The MIT Press.
- Anderson, B. A., Kim, H., Kim, A. J., Liao, M.-R., Mrkonja, L., Clement, A., & Grégoire, L. (2021). The past, present, and future of selection history. *Neuroscience & Biobehavioral Reviews*, 130, 326–350. <https://doi.org/10.1016/j.neubiorev.2021.09.004>
- Anderson, S. W., Rizzo, M., Shi, Q., Uc, E. Y., & Dawson, J. D. (2005). *Cognitive abilities related to driving performance in a simulator and crashing on the road* Driving Assessment Conference. doi:10.17077/drivingassessment.1173.
- Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. In *Psychology of learning and motivation* (Vol. 2, pp. 89–195). Elsevier.
- Baddeley, A. (1992). Working memory. *Science*, 255(5044), 556–559. <https://doi.org/10.1126/science.1736359>
- Beanland, V., Filtz, A. J., & Jeans, R. (2017). Change detection in urban and rural driving scenes: Effects of target type and safety relevance on change blindness. *Accident Analysis & Prevention*, 100, 111–122. <https://doi.org/10.1016/j.aap.2017.01.011>
- Blane, A., Falkmer, T., Lee, H. C., & Dukic Willstrand, T. (2018). Investigating cognitive ability and self-reported driving performance of post-stroke adults in a driving simulator. *Topics in Stroke Rehabilitation*, 25(1), 44–53. <https://doi.org/10.1080/10749357.2017.1373929>
- Brass, M., Ruge, H., Meiran, N., Rubin, O., Koch, I., Zysset, S., Prinz, W., & von Cramon, D. Y. (2003). When the same response has different meanings: Recoding the response meaning in the lateral prefrontal cortex. *NeuroImage*, 20(2), 1026–1031. [https://doi.org/10.1016/S1053-8119\(03\)00357-4](https://doi.org/10.1016/S1053-8119(03)00357-4)
- Braunagel, C., Rosenstiel, W., & Kasneci, E. (2017). Ready for take-over? a new driver assistance system for an automated classification of driver take-over readiness. *IEEE Intelligent Transportation Systems Magazine*, 9(4), 10–22. <https://doi.org/10.1109/ITS.2017.2743165>
- Broadbent, D. P., D'Innocenzo, G., Ellmers, T. J., Parsler, J., Szameitat, A. J., & Bishop, D. T. (2023). Cognitive load, working memory capacity and driving performance: A preliminary fNIRS and eye tracking study. *Transportation Research Part F: Traffic Psychology and Behaviour*, 92, 121–132. <https://doi.org/10.1016/j.trf.2022.11.013>
- Brookhuis, K. A., van Driel, C. J. G., Hof, T., van Arem, B., & Hoedemaeker, M. (2009). Driving with a congestion assistant; mental workload and acceptance. *Applied Ergonomics*, 40(6), 1019–1025. <https://doi.org/10.1016/j.apergo.2008.06.010>
- Caird, J. K., Simmons, S. M., Wiley, K., Johnston, K. A., & Horrey, W. J. (2018). Does Talking on a Cell Phone, With a Passenger, or Dialing Affect Driving Performance? An Updated Systematic Review and Meta-Analysis of Experimental Studies. *Human factors*, 60(1), 101–133. <https://doi.org/10.1177/0018720817748145>
- Boudreau, A., & Strybel, T. Z. (2022). The Simon Effect in Driving. *HCI International 2022*. doi:10.1007/978-3-031-17615-9_43.
- CDC. (2022). *Distracted Driving*. U.S. Department of Health & Human Services. https://www.cdc.gov/transportationsafety/distracted_driving/index.html.
- Chai, C., Lei, Y., Wei, H., Wu, C., Zhang, W., Hansen, P., Fan, H., & Shi, J. (2024). The effects of various auditory takeover requests: A simulated driving study considering the modality of non-driving-related tasks. *Applied Ergonomics*, 118, Article 104252. <https://doi.org/10.1016/j.apergo.2024.104252>
- Chase, W. G., & Ericsson, K. A. (1982). Skill and working memory. In *Psychology of learning and motivation* (Vol. 16, pp. 1–58). Elsevier.
- Chen, F., Lu, G., Lin, Q., Zhai, J., & Tan, H. (2021). Are novice drivers competent to take over control from level 3 automated vehicles? A comparative study with experienced drivers. *Transportation Research Part F: Traffic Psychology and Behaviour*, 81, 65–81. <https://doi.org/10.1016/j.trf.2021.05.012>
- Choi, D., Sato, T., Ando, T., Abe, T., Akamatsu, M., & Kitazaki, S. (2020). Effects of cognitive and visual loads on driving performance after take-over request (TOR) in automated driving. *Applied Ergonomics*, 85, Article 103074. <https://doi.org/10.1016/j.apergo.2020.103074>
- Cohen, R. A. (2014). *The neuropsychology of attention* (2nd ed.). Springer Science + Business Media. <https://doi.org/10.1007/978-0-387-72639-7>
- Cooper, J. M., Medeiros-Ward, N., & Strayer, D. L. (2013). The impact of eye movements and cognitive workload on lateral position variability in driving. *Human factors*, 55(5), 1001–1014. <https://doi.org/10.1177/0018720813480177>

- Deng, C., Cao, S., Wu, C., & Lyu, N. (2019). Modeling driver take-over reaction time and emergency response time using an integrated cognitive architecture. *Transportation research record*, 2673(12), 380–390. <https://doi.org/10.1177/0361198119842114>
- Depestele, S., Ross, V., Verstraeten, S., Brijis, K., Brijis, T., van Dun, K., & Meesen, R. (2020). The impact of cognitive functioning on driving performance of older persons in comparison to younger age groups: A systematic review. *Transportation Research Part F: Traffic Psychology and Behaviour*, 73, 433–452. <https://doi.org/10.1016/j.trf.2020.07.009>
- Devlin, A., McGillivray, J., Charlton, J., Lowndes, G., & Etienne, V. (2012). Investigating driving behaviour of older drivers with mild cognitive impairment using a portable driving simulator. *Accident Analysis & Prevention*, 49, 300–307. <https://doi.org/10.1016/j.aap.2012.02.022>
- Chu, D., Wang, R., Deng, Y., Lu, L., & Wu, C. (2020). *Vibrotactile Take-over Requests in Highly Automated Driving* 2020 4th CAA International Conference on Vehicular Control and Intelligence (CVCI), Hangzhou, China. doi:10.1109/CVCI51460.2020.9338667.
- Dobbs, B. M., Wodzin, E., & Vegega, M. (2005). *Medical Conditions and Driving: A Review of the Literature (1960-2000)*. (DOT HS 809 690). Retrieved from <https://rosap.nhtl.bts.gov/view/dot/1902>.
- Drummond, A. E. (1989). *An overview of novice driver performance issues: A literature review*. Monash University Accident Research Centre.
- Du, N., Kim, J., Zhou, F., Pulver, E., Tilbury, D. M., Robert, L. P., Pradhan, A. K., & Yang, X. J. (2020). Evaluating Effects of Cognitive Load, Takeover Request Lead Time, and Traffic Density on Drivers'. In *Takeover Performance in Conditionally Automated Driving 12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications, Virtual Event*. <https://doi.org/10.1145/3409120.3410666>
- Du, N., Yang, X. J., & Zhou, F. (2020). Psychophysiological responses to takeover requests in conditionally automated driving. *Accident Analysis & Prevention*, 148, Article 105804. <https://doi.org/10.1016/j.aap.2020.105804>
- Eich, T. S., Langfield, C., Sakhardande, J., Gazes, Y., Habeck, C., & Stern, Y. (2023). Older adults compensate for switch, but not mixing costs, relative to younger adults on an intrinsically cued task switching experiment [Original Research]. *Frontiers in Aging Neuroscience*, 15. <https://doi.org/10.3389/fnagi.2023.1152582>
- Eriksson, A., & Stanton, N. A. (2017). Takeover time in highly automated vehicles: noncritical transitions to and from manual control. *Human factors*, 59(4), 689–705. <https://doi.org/10.1177/0018720816685832>
- Eysenck, M. W., & Keane, M. T. (2020). Cognitive psychology: A student's handbook. *Psychology press*. <https://doi.org/10.4324/9781351058513>
- Feldhütter, A., Gold, C., Schneider, S., & Bengler, K. (2017). *How the Duration of Automated Driving Influences Take-Over Performance and Gaze Behavior Advances in Ergonomic Design of Systems*. Berlin, Heidelberg: Products and Processes.
- Furley, P. A., & Memmert, D. (2012). Working memory capacity as controlled attention in tactical decision making. *Journal of Sport and Exercise Psychology*, 34(3), 322–344. <https://doi.org/10.1123/jsep.34.3.322>
- Gajewski, P. D., Hanisch, E., Falkenstein, M., Thönes, S., & Wascher, E. (2018). What does the n-back task measure as we get older? relations between working-memory measures and other cognitive functions across the lifespan [original research]. *Frontiers in Psychology*, 9. <https://doi.org/10.3389/fpsyg.2018.02208>
- Gamboz, N., Borella, E., & Brandimonte, M. A. (2009). The role of switching, inhibition and working memory in older adults' performance in the wisconsin card sorting test. *Aging, Neuropsychology, and Cognition*, 16(3), 260–284. <https://doi.org/10.1080/13825580802573045>
- Garrison, H., Scholey, A., Ogden, E., & Benson, S. (2021). The effects of alcohol intoxication on cognitive functions critical for driving: A systematic review. *Accident Analysis & Prevention*, 154, Article 106052. <https://doi.org/10.1016/j.aap.2021.106052>
- Giorgetti, R., Marcotulli, A., Tagliabacci, A., & Schifano, F. (2015). Effects of ketamine on psychomotor, sensory and cognitive functions relevant for driving ability. *Forensic Science International*, 252, 127–142. <https://doi.org/10.1016/j.forsciint.2015.04.024>
- Gong, J., Guo, X., Qi, C., Liang, X., & Wang, Q. (2023). Measuring takeover performance in different driving scenarios: Considering the influence of non-driving-related tasks and takeover request lead time. *Transportation Research Part F: Traffic Psychology and Behaviour*, 97, 411–421. <https://doi.org/10.1016/j.trf.2023.08.002>
- Goodridge, C. M., Gonçalves, R. C., & Öztürk, İ. (2023). *What do we mean by cognitive load? Towards more accurate definition of the term for better identification by driver monitoring systems Adjunct*. In *Proceedings of the 15th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. <https://doi.org/10.1145/3581961.3609839>
- Guettas, A., Ayad, S., & Kazar, O. (2020). Driver State Monitoring System. In *A Review Proceedings of the 4th International Conference on Big Data and Internet of Things*. <https://doi.org/10.1145/3372938.3372966>
- Hamish Janson, A., & Merat, N. (2005). Surrogate in-vehicle information systems and driver behaviour: Effects of visual and cognitive load in simulated rural driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 8(2), 79–96. <https://doi.org/10.1016/j.trf.2005.04.002>
- Hautekiet, C., Verschooren, S., Langerock, N., & Vergauwe, E. (2023). Attentional switching between perception and memory: Examining asymmetrical switch costs. *Attention, Perception, & Psychophysics*, 85(5), 1398–1408. <https://doi.org/10.3758/s13414-023-02665-w>
- Hecht, T., Feldhütter, A., Radlmayr, J., Nakano, Y., Miki, Y., Henle, C., & Bengler, K. (2019, 2019). A Review of Driver State Monitoring Systems in the Context of Automated Driving. Proceedings of the 20th Congress of the International Ergonomics Association (IEA 2018), Cham.
- Hecht, T., Kratzert, S., & Bengler, K. (2020). The effects of a predictive HMI and different transition frequencies on acceptance, workload, usability, and gaze behavior during urban automated driving. *Information*, 11(2), 73. <https://doi.org/10.3390/info11020073>
- Hiebel, N., & Zimmer, H. D. (2015). Individual differences in working memory capacity and attentional control. *Canadian Journal of Experimental Psychology / Revue canadienne de psychologie expérimentale*, 69(1), 17–27. <https://doi.org/10.1037/cep0000029>
- Hong, S., & Yang, J. H. (2022). Effect of multimodal takeover request issued through A-pillar LED light, earcon, speech message, and haptic seat in conditionally automated driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 89, 488–500. <https://doi.org/10.1016/j.trf.2022.07.012>
- Huang, J., Zhang, Q., Zhang, T., Wang, T., & Tao, D. (2024). Assessment of drivers' mental workload by multimodal measures during auditory-based dual-task driving scenarios. *Sensors*, 24(3), 1041. <https://doi.org/10.3390/s24031041>
- Hungund, A. P., & Kumar Pradhan, A. (2023). Impact of non-driving related tasks while operating automated driving systems (ADS): A systematic review. *Accident Analysis & Prevention*, 188, Article 107076. <https://doi.org/10.1016/j.aap.2023.107076>
- Hurts, K. (2011). Driver distraction, crosstalk, and spatial reasoning. *Transportation Research Part F: Traffic Psychology and Behaviour*, 14(4), 300–312. <https://doi.org/10.1016/j.trf.2011.02.003>
- ISO. (2010). Road vehicles-Ergonomic aspects of transport information and control systems-Simulated lane change test to assess in-vehicle secondary task demand. In (Vol. ISO/TS 26022:2010(E)). Switzerland: the International Organization for Standardization.
- ISO. (2019). Road vehicles-Ergonomic aspects of transport information and control systems-Calibration tasks for methods which assess driver demand due to the use of in-vehicle systems. In (Vol. ISO/TS 14198:2019(E)). Switzerland: the International Organization for Standardization.
- Jaussein, M., Léveque, L., Deniel, J., Bellet, T., Tattgrain, H., & Marin-Lamellet, C. (2021). How do non-driving-related tasks affect engagement under highly automated driving situations? a literature review [Review]. *Frontiers in Future Transportation*, 2. <https://doi.org/10.3389/ffutr.2021.687602>
- Kane, M. J., Conway, A. R., Miura, T. K., & Colflesh, G. J. (2007). Working memory, attention control, and the N-back task: A question of construct validity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 33(3), 615.
- Kane, M. J., & Engle, R. W. (2002). The role of prefrontal cortex in working-memory capacity, executive attention, and general fluid intelligence: An individual-differences perspective. *Psychonomic Bulletin & Review*, 9(4), 637–671. <https://doi.org/10.3758/BF03196323>
- Kaul, R., & Jipp, M. (2023). Influence of cognitive processes on driver decision-making in dilemma zone. *Transportation Research Interdisciplinary Perspectives*, 19, Article 100805. <https://doi.org/10.1016/j.trip.2023.100805>
- Kiesel, A., Steinhäuser, M., Wendt, M., Falkenstein, M., Jost, K., Philipp, A. M., & Koch, I. (2010). Control and interference in task switching-a review [Article]. *Psychological bulletin*, 136(5), 849–874. <https://doi.org/10.1037/a0019842>
- Kim, H., Kim, W., Kim, J., Lee, S.-J., Yoon, D., & Jo, J. (2021). A study on Re-engagement and stabilization time on take-over transition in a highly automated driving system. *Electronics*, 10(3), 344. <https://doi.org/10.3390/electronics10030344>
- Kimberg, D. Y., Aguirre, G. K., & D'Esposito, M. (2000). Modulation of task-related neural activity in task-switching: An fMRI study [Article]. *Cognitive Brain Research*, 10(1–2), 189–196. [https://doi.org/10.1016/S0926-6410\(00\)00016-1](https://doi.org/10.1016/S0926-6410(00)00016-1)

- Kotlyusov, A. I., Kasanov, D., Kosachenko, A. I., Gashkova, A. S., Pavlov, Y. G., & Malykh, S. (2023). Working memory capacity depends on attention control, but not selective attention. *Behavioral Sciences*, 13(2), 92. <https://doi.org/10.3390/bs13020092>
- Kreitz, C., Furlay, P., Memmert, D., & Simons, D. J. (2015). Inattention blindness and individual differences in cognitive abilities. *PLoS One*, 10(8), Article e0134675. <https://doi.org/10.1371/journal.pone.0134675>
- Lee, J. Y., Lee, J. D., Bärgrman, J., Lee, J., & Reimer, B. (2018). How safe is tuning a radio?: Using the radio tuning task as a benchmark for distracted driving. *Accident Analysis & Prevention*, 110, 29–37. <https://doi.org/10.1016/j.aap.2017.10.009>
- Lew, H. L., Rosen, P. N., Thomander, D., & Poole, J. H. (2009). The Potential Utility of Driving Simulators in the Cognitive Rehabilitation of Combat-Returnees With Traumatic Brain Injury. *The Journal of Head Trauma Rehabilitation*, 24(1), 51–56. <https://doi.org/10.1097/HTR.0b013e3181956fe3>
- Liang, K., Calvert, S., Nordhoff, S., & Lint, H. v. (2024, 2–5 June 2024). Examining the Role of Driver Perception in Takeover Time: Application of Task-Capability Interface Theory. 2024 IEEE Intelligent Vehicles Symposium (IV).
- Liang, N., Lim, C., Yu, D., Prakah-Asante, K. O., Curry, R., Blommer, M., Swaminathan, R., & Pitts, B. J. (2022). *The effects of non-driving-related task engagement on automated driving takeover performance during the nighttime* Human Factors and Ergonomics Society Annual Meeting, Atlanta, Georgia. doi:10.1177/1071181322661.
- Lin, Q.-F., Lyu, Y., Zhang, K.-F., & Ma, X.-W. (2021). Effects of non-driving related tasks on readiness to take over control in conditionally automated driving. *Traffic Injury Prevention*, 22(8), 629–633. <https://doi.org/10.1080/15389588.2021.1969373>
- Llerena, L. E., Aronow, K. V., Macleod, J., Bard, M., Salzman, S., Greene, W., Haider, A., & Schupper, A. (2015). An evidence-based review: Distracted driver. *Journal of Trauma and Acute Care Surgery*, 78(1), 147–152. <https://doi.org/10.1097/ta.0000000000000487>
- Lu, G., Zhai, J., Li, P., Chen, F., & Liang, L. (2021). Measuring drivers' takeover performance in varying levels of automation: Considering the influence of cognitive secondary task. *Transportation Research Part F: Traffic Psychology and Behaviour*, 82, 96–110. <https://doi.org/10.1016/j.trf.2021.08.005>
- Ma, J., Gong, Z., Tan, J., Zhang, Q., & Zuo, Y. (2020). Assessing the driving distraction effect of vehicle HMI displays using data mining techniques. *Transportation Research Part F: Traffic Psychology and Behaviour*, 69, 235–250. <https://doi.org/10.1016/j.trf.2020.01.016>
- Ma, J., Li, J., Wang, W., Huang, H., Zhang, X., & Zhao, J. (2024). Exploring the impact of the passenger's display on driver workload and driving performance. *Applied Ergonomics*, 114, Article 104138. <https://doi.org/10.1016/j.apergo.2023.104138>
- MacPherson, S. E. (2018). Definition: Dual-tasking and multitasking. *Cortex*, 106, 313–314.
- Madore, K. P., & Wagner, A. D. (2019). Multicosts of Multitasking. *Cerebrum*, 2019.
- Mamo, W. G., Alhajjaseen, W. K. M., Brijs, K., Dirix, H., Vanroelen, G., Hussain, Q., Brijs, T., & Ross, V. (2024). The impact of cognitive load on a lane change task (LCT) among male autistic individuals: A driving simulator study. *Transportation Research Part F: Traffic Psychology and Behaviour*, 106, 27–43. <https://doi.org/10.1016/j.trf.2024.07.030>
- Martens, M. H., & Brouwer, R. F. T. (2013). Measuring being lost in thought: An exploratory driving simulator study. *Transportation Research Part F: Traffic Psychology and Behaviour*, 20, 17–28. <https://doi.org/10.1016/j.trf.2013.04.002>
- McDonald, A. D., Alambeigi, H., Engström, J., Markkula, G., Vogelphol, T., Dunne, J., & Yuma, N. (2019). Toward Computational Simulations of Behavior During Automated Driving Takeovers: A Review of the Empirical and Modeling Literatures. *Human factors*, 61(4), 642–688. <https://doi.org/10.1177/0018720819829572>
- McKendrick, R., & Harwood, A. (2019). Cognitive Workload and Workload Transitions Elicit Curvilinear Hemodynamics During Spatial Working Memory [Original Research]. *Frontiers in Human Neuroscience*, 13. <https://doi.org/10.3389/fnhum.2019.00405>
- McWilliams, T., & Ward, N. (2021). Underload on the Road: Measuring Vigilance Decrements During Partially Automated Driving [Review]. *Frontiers in Psychology*, 12. <https://doi.org/10.3389/fpsyg.2021.631364>
- Melnick, V., Thompson, S., Jennings, P., & Birrell, S. (2021). Effect of cognitive load on drivers' State and task performance during automated driving: Introducing a novel method for determining stabilisation time following take-over of control. *Accident Analysis & Prevention*, 151, Article 105967. <https://doi.org/10.1016/j.aap.2020.105967>
- Mercedes-Benz. (2023). *Certification for SAE Level 3 system for U.S. market*. Mercedes-Benz Group. <https://group.mercedes-benz.com/innovation/product-innovation/autonomous-driving/drive-pilot-nevada.html>
- Merat, N., Jamson, A. H., Lai, F. C. H., Daly, M., & Carsten, O. M. J. (2014). Transition to manual: Driver behaviour when resuming control from a highly automated vehicle. *Transportation Research Part F: Traffic Psychology and Behaviour*, 27, 274–282. <https://doi.org/10.1016/j.trf.2014.09.005>
- Meteier, Q., De Salis, E., Capallera, M., Widmer, M., Angelini, L., Abou Khaled, O., Sonderegger, A., & Mugellini, E. (2022). Relevant physiological indicators for assessing workload in conditionally automated driving, through three-class classification and regression. *Frontiers in Computer Science*, 3, Article 775282. <https://doi.org/10.3389/fcomp.2021.775282>
- Mochi, F., & Madjar, N. (2018). Chapter 5 - Interruptions and Multitasking: Advantages and Disadvantages for Creativity at Work. In R. Reiter-Palmon, V. L. Kennel, & J. C. Kaufman (Eds.), *Individual Creativity in the Workplace* (pp. 103–127). Academic Press. <https://doi.org/10.1016/B978-0-12-813238-8.00005-X>
- Monsell, S. (2003). Task switching. *Trends in Cognitive Sciences*, 7(3), 134–140. [https://doi.org/10.1016/S1364-6613\(03\)00028-7](https://doi.org/10.1016/S1364-6613(03)00028-7)
- Monsell, S., Sumner, P., & Waters, H. (2003). Task-set reconfiguration with predictable and unpredictable task switches. *Memory & Cognition*, 31(3), 327–342. <https://doi.org/10.3758/BF03194391>
- Morales-Alvarez, W., Sipele, O., Léberon, R., Tadjine, H. H., & Olaverri-Monreal, C. (2020). Automated driving: A literature review of the take over request in conditional automation. *Electronics*, 9(12), 2087. <https://doi.org/10.3390/electronics9122087>
- Naujoks, F., Purucker, C., Wiedemann, K., & Marberger, C. (2019). Noncritical state transitions during conditionally automated driving on german freeways: effects of non-driving related tasks on takeover time and takeover quality. *Human factors*, 61(4), 596–613. <https://doi.org/10.1177/0018720818824002>
- NHTSA. (2012). *Visual-manual NHTSA driver distraction guidelines for in-vehicle electronic devices*. (2013-09883). Washington, DC: National Highway Traffic Safety Administration Retrieved from <https://www.federalregister.gov/documents/2013/04/26/2013-09883/visual-manual-nhtsa-driver-distraction-guidelines-for-in-vehicle-electronic-devices#print>
- Oberauer, K. (2019). Working memory and attention – a conceptual analysis and review. *Journal of Cognition*. <https://doi.org/10.5334/joc.58>
- Oliver, N., Calvard, T., & Potočník, K. (2017). The tragic crash of Flight AF447 shows the unlikely but catastrophic consequences of automation. *Harvard Business Review*, 15. <https://hbr.org/2017/09/the-tragic-crash-of-flight-af447-shows-the-unlikely-but-catastrophic-consequences-of-automation>
- Overton, T. L., Rives, T. E., Hecht, C., Shafi, S., & Gandhi, R. R. (2015). Distracted driving: Prevalence, problems, and prevention. *International Journal of Injury Control and Safety Promotion*, 22(3), 187–192. <https://doi.org/10.1080/17457300.2013.879482>
- Oviedo-Trespalacios, O., Haque, M. M., King, M., & Washington, S. (2016). Understanding the impacts of mobile phone distraction on driving performance: A systematic review. *Transportation Research Part C: Emerging Technologies*, 72, 360–380. <https://doi.org/10.1016/j.trc.2016.10.006>
- Öztürk, İ., Merat, N., Rowe, R., & Fotios, S. (2023). The effect of cognitive load on Detection-Response Task (DRT) performance during day- and night-time driving: A driving simulator study with young and older drivers. *Transportation Research Part F: Traffic Psychology and Behaviour*, 97, 155–169. <https://doi.org/10.1016/j.trf.2023.07.002>
- Parasuraman, R., & Caggiano, D. (2002). Mental Workload. In V. S. Ramachandran (Ed.), *Encyclopedia of the Human Brain* (pp. 17–27). Academic Press. Doi: 10.1016/B0-12-227210-2/00206-5.
- Pashler, H. (1994). Dual-task interference in simple tasks: Data and theory. *Psychological bulletin*, 116(2), 220.
- Pashler, H. (2000). Task switching and multitask performance. In S. Monsell & J. Driver (Eds.), *Attention and Performance XVIII: Control of cognitive processes* (pp. 275–307). MIT Press.
- Patten, C. J. D., Kircher, A., Östlund, J., Nilsson, L., & Svenson, O. (2006). Driver experience and cognitive workload in different traffic environments. *Accident Analysis & Prevention*, 38(5), 887–894. <https://doi.org/10.1016/j.aap.2006.02.014>
- Peng, Q., Wu, Y., Qie, N., & Iwaki, S. (2022). Age-related effects of executive function on takeover performance in automated driving. *Scientific reports*, 12(1), 5410. <https://doi.org/10.1038/s41598-022-08522-4>

- Peng, Q., Wu, Y., Sato, T., & Iwaki, S. (2023). *The Relationship Between Older Drivers' Cognitive Ability and Takeover Performance in Conditionally Automated Driving*. Human Aspects of IT for the Aged Population, Cham. https://link.springer.com/chapter/10.1007/978-3-031-34866-2_8.
- Petermeijer, S. M., Cieler, S., & de Winter, J. C. F. (2017). Comparing spatially static and dynamic vibrotactile take-over requests in the driver seat. *Accident Analysis & Prevention*, 99, 218–227. <https://doi.org/10.1016/j.aap.2016.12.001>
- Petzoldt, T., Bellem, H., & Krems, J. F. (2014). The critical tracking task: A potentially useful method to assess driver distraction? *Hum Factors*, 56(4), 789–808. <https://doi.org/10.1177/0018720813501864>
- Radlmayr, J., Gold, C., Lorenz, L., Farid, M., & Bengler, K. (2014). How traffic situations and non-driving related tasks affect the take-over quality in highly automated driving. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 58(1), 2063–2067. <https://doi.org/10.1177/1541931214581434>
- Ranchet, M., Broussolle, E., Poisson, A., & Paire-Ficout, L. (2012). Relationships between cognitive functions and driving behavior in parkinson's disease. *European Neurology*, 68(2), 98–107. <https://doi.org/10.1159/000338264>
- Regan, M. A., Hallett, C., & Gordon, C. P. (2011). Driver distraction and driver inattention: Definition, relationship and taxonomy. *Accident Analysis & Prevention*, 43(5), 1771–1781. <https://doi.org/10.1016/j.aap.2011.04.008>
- Remington, R. W., & Loft, S. (2015). Attention and multitasking. In *APA handbook of human systems integration*. (pp. 261–276). American Psychological Association. .
- Rensink, R. A. (2009). Change Blindness and Inattention Blindness. In W. Banks (Ed.), *Encyclopedia of Consciousness*, vol 1 (pp. 47–59). Elsevier.
- Robison, M. K., & Unsworth, N. (2017). Individual differences in working memory capacity predict learned control over attentional capture. *Journal of Experimental Psychology: Human Perception and Performance*, 43(11), 1912–1924. <https://doi.org/10.1037/xhp0000419>
- Rogers, R. D., & Monsell, S. (1995). Costs of a predictable switch between simple cognitive tasks. *Journal of Experimental Psychology: General*, 124(2), 207–231. <https://doi.org/10.1037/0096-3445.124.2.207>
- Romero-Ayuso, D., Castellero-Perea, A., González, P., Navarro, E., Molina-Massó, J. P., Funes, M. J., Ariza-Vega, P., Toledano-González, A., & Triviño-Juárez, J. M. (2021). Assessment of cognitive instrumental activities of daily living: A systematic review. *Disabil Rehabil*, 43(10), 1342–1358. <https://doi.org/10.1080/09638288.2019.1665720>
- Ronca, V., Brambati, F., Napoletano, L., Marx, C., Trösterer, S., Vozzi, A., Aricò, P., Giorgi, A., Capotorto, R., Borghini, G., Babiloni, F., & Di Flumeri, G. (2024). A Novel EEG-based assessment of distraction in simulated driving under different road and traffic conditions. *Brain Sciences*, 14(3), 193. <https://doi.org/10.3390/brainsci14030193>
- SAE. (2021). Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles. In (Vol. J3016_202104): SAE International.
- Schneider, D., & Logan, G. (2009). Task switching. *Encyclopedia of neuroscience*, 9, 869–874.
- Schneider, W., & Shiffrin, R. M. (1977). Controlled and automatic human information processing: I. Detection, search, and attention. *Psychological Review*, 84(1), 1–66. <https://doi.org/10.1037/0033-295X.84.1.1>
- Servant, M., Cassey, P., Woodman, G. F., & Logan, G. D. (2018). Neural bases of automaticity. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 44(3), 440–464. <https://doi.org/10.1037/xlm0000454>
- Shi, E., & Bengler, K. (2022a). Non-driving related tasks' effects on takeover and manual driving behavior in a real driving setting: A differentiation approach based on task switching and modality shifting. *Accident Analysis & Prevention*, 178, Article 106844. <https://doi.org/10.1016/j.aap.2022.106844>
- Shi, E., & Bengler, K. (2022b). Using Task Switching to Explain Effects of Non-Driving Related Activities on Takeover and Manual Driving Behavior Following Level 3 Automated Driving Adjunct Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications, Seoul Korea.
- Shiffrin, R. M., & Schneider, W. (1977). Controlled and automatic human information processing: II. Perceptual learning, automatic attending and a general theory. *Psychological Review*, 84(2), 127–190. <https://doi.org/10.1037/0033-295X.84.2.127>
- Shimonaga, K., Hama, S., Tsuji, T., Yoshimura, K., Nishino, S., Yanagawa, A., Soh, Z., Matsushige, T., Mizoue, T., Onoda, K., Yamashita, H., Yamawaki, S., & Kurisu, K. (2021). The right hemisphere is important for driving-related cognitive function after stroke. *Neurosurgical Review*, 44(2), 977–985. <https://doi.org/10.1007/s10143-020-01272-9>
- Siéroff, E. (2001). Attention: Multiple Resources. In N. J. Smelser & P. B. Baltes (Eds.), *International Encyclopedia of the Social & Behavioral Sciences* (pp. 884–888). Pergamon. Doi: 10.1016/B0-08-043076-7/01446-7.
- Skaugset, L. M., Farrell, S., Carney, M., Wolff, M., Santen, S. A., Perry, M., & Cico, S. J. (2016). Can you multitask? Evidence and limitations of task switching and multitasking in emergency medicine. *Annals of Emergency Medicine*, 68(2), 189–195. <https://doi.org/10.1016/j.annemergmed.2015.10.003>
- Sohn, M. H., Ursu, S., Anderson, J. R., Stenger, V. A., & Carter, C. S. (2000). The role of prefrontal cortex and posterior parietal cortex in task switching [Article]. *Proceedings of the National Academy of Sciences of the United States of America*, 97(24), 13448–13453. <https://doi.org/10.1073/pnas.240460497>
- Solso, R. L., MacLin, M. K., & MacLin, O. H. (2005). *Cognitive psychology* (7th ed.). Pearson Education New Zealand.
- Stojmenova, K., Jakus, G., & Sodnik, J. (2017). Sensitivity evaluation of the visual, tactile, and auditory detection response task method while driving. *Traffic Injury Prevention*, 18(4), 431–436. <https://doi.org/10.1080/15389588.2016.1214868>
- Stojmenova, K., & Sodnik, J. (2018). Detection-response task-uses and limitations. *Sensors (Basel)*, 18(2). <https://doi.org/10.3390/s18020594>
- Strobach, T., Liepelt, R., Schubert, T., & Kiesel, A. (2012). Task switching: Effects of practice on switch and mixing costs. *Psychological Research*, 76(1), 74–83. <https://doi.org/10.1007/s00426-011-0323-x>
- Strobach, T., Wendt, M., & Janczyk, M. (2018). Editorial: multitasking: Executive functioning in dual-task and task switching situations [Editorial]. *Frontiers in Psychology*, 9. <https://doi.org/10.3389/fpsyg.2018.00108>
- Sweller, J. (2011). Chapter two - Cognitive Load Theory. In J. P. Mestre, & B. H. Ross (Eds.), *Psychology of Learning and Motivation* (Vol. 55, pp. 37–76). Academic Press. <https://doi.org/10.1016/B978-0-12-387691-1.00002-8>.
- Sweller, J., & Sweller, S. (2006). Natural information processing systems. *Evolutionary Psychology*, 4(1), 434–458. <https://doi.org/10.1177/147470490600400135>
- Ucińska, M., Odachowska, E., Gąsiorek, K., & Kruszewski, M. (2021). Age and experience in driving a vehicle and psychomotor skills in the context of automation. *Open Engineering*, 11(1), 453–462. <https://doi.org/10.1515/eng-2021-0045>
- UNECE. (2021). Concerning the Adoption of Harmonized Technical United Nations Regulations for Wheeled Vehicles, Equipment and Parts which can be Fitted and/or be Used on Wheeled Vehicles and the Conditions for Reciprocal Recognition of Approvals Granted on the Basis of these United Nations Regulations. In (Vol. UN Regulation No. 157). Geneva: United Nations Economic Commission for Europe.
- van Winsum, W. (2018). The effects of cognitive and visual workload on peripheral detection in the detection response task. *Human factors*, 60(6), 855–869. <https://doi.org/10.1177/0018720818776880>
- Vance, D. E., Fazeli, P. L., Ball, D. A., Slater, L. Z., & Ross, L. A. (2014). Cognitive functioning and driving simulator performance in middle-aged and older adults With HIV. *Journal of the Association of Nurses in AIDS Care*, 25(2), e11–e26. <https://doi.org/10.1016/j.jana.2013.12.001>
- VicRoads. (2023a, 21 Sept 2023). *Mobile phones, technology & driving*. Victoria Government. <https://www.vicroads.vic.gov.au/safety-and-road-rules/driver-safety/mobile-phones-and-driving>.
- VicRoads. (2023b). *Reducing P driver crash risk*. Victoria State Government. <https://www.vicroads.vic.gov.au/safety-and-road-rules/driver-safety/young-and-new-drivers/reducing-p-driver-crash-risk>.
- Voinea, G.-D., Boboc, R. G., Buzdugan, I.-D., Antonya, C., & Yannis, G. (2023). Texting while driving: a literature review on driving simulator studies. *International Journal of Environmental Research and Public Health*, 20(5), 4354. <https://doi.org/10.3390/ijerph20054354>
- von Janczewski, N., Wittmann, J., Engeln, A., Baumann, M., & Krauß, L. (2021). A meta-analysis of the n-back task while driving and its effects on cognitive workload. *Transportation Research Part F: Traffic Psychology and Behaviour*, 76, 269–285. <https://doi.org/10.1016/j.trf.2020.11.014>
- Waszak, F., Hommel, B., & Allport, A. (2003). Task-switching and long-term priming: Role of episodic stimulus–task bindings in task-shift costs. *Cognitive Psychology*, 46(4), 361–413. [https://doi.org/10.1016/S0010-0285\(02\)00520-0](https://doi.org/10.1016/S0010-0285(02)00520-0)
- Wechsler, K., Drescher, U., Janouch, C., Haeger, M., Voelcker-Rehage, C., & Bock, O. (2018). Multitasking during simulated car driving: a comparison of young and older persons [original research]. *Frontiers in Psychology*, 9. <https://doi.org/10.3389/fpsyg.2018.00910>
- Wickens, C. D. (2008). Multiple resources and mental workload. *Hum Factors*, 50(3), 449–455. <https://doi.org/10.1518/001872008x288394>

- Wickens, C. D. (2014). The structure of attentional resources. In *Attention and performance VIII* (pp. 239–257). Psychology Press. <https://doi.org/10.4324/9781315802961>.
- Wiedemann, K., Naujoks, F., Wörle, J., Kenntner-Mabiala, R., Kaussner, Y., & Neukum, A. (2018). Effect of different alcohol levels on take-over performance in conditionally automated driving. *Accident Analysis & Prevention*, 115, 89–97. <https://doi.org/10.1016/j.aap.2018.03.001>
- Wu, C., Wu, H., Lyu, N., & Zheng, M. (2019). Take-over performance and safety analysis under different scenarios and secondary tasks in conditionally automated driving. *IEEE Access*, 7, 136924–136933. <https://doi.org/10.1109/ACCESS.2019.2914864>
- Wu, Y., Kihara, K., Takeda, Y., Sato, T., Akamatsu, M., & Kitazaki, S. (2019). Effects of scheduled manual driving on drowsiness and response to take over request: A simulator study towards understanding drivers in automated driving. *Accident Analysis & Prevention*, 124, 202–209. <https://doi.org/10.1016/j.aap.2019.01.013>
- Xu, J., Fard, M., Zhang, N., Davy, J. L., & Robinson, S. R. (2024). Interrelatedness of steering and lateral position parameters: Recommendations for the assessment of driving performance. *Journal of safety research*, 88, 275–284. <https://doi.org/10.1016/j.jsr.2023.11.014>
- Xue, Q., Wang, X., Li, Y., & Guo, W. (2023). Young novice drivers' cognitive distraction detection: comparing support vector machines and random forest model of vehicle control behavior. *Sensors*, 23(3), 1345. <https://doi.org/10.3390/s23031345>
- Yang, B., Inoue, K., Yan, Z., Wang, Z., Kitazaki, S., & Nakano, K. (2023). Influences of Level 2 Automated Driving on Driver Behaviors: A Comparison With Manual Driving. *IEEE transactions on intelligent transportation systems*, 1–15. <https://doi.org/10.1109/TITS.2023.3308569>
- Yang, J., Liang, N., Pitts, B. J., Prakah-Asante, K., Curry, R., & Yu, D. (2023). An eye-fixation related electroencephalography technique for predicting situation awareness: implications for driver state monitoring systems. *Human factors*. <https://doi.org/10.1177/00187208231204570>, 00187208231204570.
- Yang, Y., Ye, Z., Easa, S. M., Feng, Y., & Zheng, X. (2023). Effect of driving distractions on driver mental workload in work zone's warning area. *Transportation Research Part F: Traffic Psychology and Behaviour*, 95, 112–128. <https://doi.org/10.1016/j.trf.2023.03.018>
- Yeung, N., & Monsell, S. (2003). Switching between tasks of unequal familiarity: The role of stimulus-attribute and response-set selection. *Journal of Experimental Psychology: Human Perception and Performance*, 29(2), 455–469. <https://doi.org/10.1037/0096-1523.29.2.455>
- Young, M. S., & Stanton, N. A. (2002a). Attention and automation: New perspectives on mental underload and performance. *Theoretical Issues in Ergonomics Science*, 3(2), 178–194. <https://doi.org/10.1080/14639220210123789>
- Young, M. S., & Stanton, N. A. (2002b). Malleable attentional resources theory: a new explanation for the effects of mental underload on performance. *Human factors*, 44(3), 365–375. <https://doi.org/10.1518/0018720024497709>
- Yu, D., Park, C., Choi, H., Kim, D., & Hwang, S.-H. (2021). Takeover safety analysis with driver monitoring systems and driver-vehicle interfaces in highly automated vehicles. *Applied Sciences*, 11(15), 6685. <https://doi.org/10.3390/app11156685>
- Zeeb, K., Buchner, A., & Schrauf, M. (2015). What determines the take-over time? An integrated model approach of driver take-over after automated driving. *Accident Analysis & Prevention*, 78, 212–221. <https://doi.org/10.1016/j.aap.2015.02.023>
- Zeeb, K., Buchner, A., & Schrauf, M. (2016). Is take-over time all that matters? The impact of visual-cognitive load on driver take-over quality after conditionally automated driving. *Accident Analysis & Prevention*, 92, 230–239. <https://doi.org/10.1016/j.aap.2016.04.002>
- Zhai, J., Lu, G., & Chen, F. (2020). The Effects of Extra Cognitive Workload on Drivers' Driving Performance under Smooth Car-Following Drive and Critical Situations. In *International Conference on Transportation and Development* (pp. 289–300). <https://doi.org/10.1061/9780784483145.026>
- Zhang, H., Guo, Y., Yuan, W., & Li, K. (2023). On the importance of working memory in the driving safety field: A systematic review. *Accident Analysis & Prevention*, 187, Article 107071. <https://doi.org/10.1016/j.aap.2023.107071>
- Zhang, N., Fard, M., Davy, J. L., Parida, S., & Robinson, S. R. (2023). Is driving experience all that matters? Drivers' takeover performance in conditionally automated driving. *Journal of safety research*, 87, 323–331. <https://doi.org/10.1016/j.jsr.2023.08.003>
- Zhang, N., Fard, M., Xu, J., Davy, J. L., & Robinson, S. R. (2023). Influence of non-driving related tasks on driving performance after takeover transition in conditionally automated driving. *Transportation Research Part F: Traffic Psychology and Behaviour*, 96, 248–264. <https://doi.org/10.1016/j.trf.2023.05.009>
- Zhou, H., Kamijo, K., Itoh, M., & Kitazaki, S. (2021). Effects of explanation-based knowledge regarding system functions and driver's roles on driver takeover during conditionally automated driving: A test track study. *Transportation Research Part F: Traffic Psychology and Behaviour*, 77, 1–9. <https://doi.org/10.1016/j.trf.2020.11.015>